

How Political Insiders Lose Out When International Aid Underperforms: Evidence from a Participatory Development Experiment in Ghana¹

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Participatory development is designed to mitigate problems of political bias in pre-existing local government but also interacts with it in complex ways. Using a five-year randomized controlled study in 97 clusters of villages (194 villages) in Ghana, we analyze the effects of a major participatory development program on participation in, leadership of and investment by pre-existing political institutions, and on households' overall socioeconomic well-being. Applying theoretical insights on political participation and redistributive politics, we consider the possibility of both cross-institutional mobilization and displacement, and heterogeneous effects by partisanship. We find the government and its political supporters acted with high expectations for the participatory approach: treatment led to increased participation in local governance and reallocation of resources. But the results did not meet expectations, resulting in a worsening of socioeconomic wellbeing in treatment versus control villages for government supporters. This demonstrates international aid's complex distributional consequences.

Keywords: participatory development; political economy; international aid; distributive politics

JEL Codes: H4; H7; O12; O17; O19

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I. Introduction

The received wisdom is that government supporters benefit most from traditional donor aid to governments (Briggs 2012; 2014; Hodler and Raschky 2014; Jablonski 2014). The phenomenon of international aid to non-governmental organizations (NGOs), especially participatory development aid, is in part a response to the problem of misdirected aid (Dietrich 2013; Mansuri and Rao 2013). Participatory development approaches aim to build new local institutional structures to administer aid, with the goal of achieving more effective projects and equitable outcomes. The expectation is that the new institutions developed through these approaches should be able to deliver aid neutrally, achieving benefits for citizens across the political spectrum.

We study the expansion of a participatory development program in Ghana, and find that it interacts with pre-existing political institutions in complex ways. For a complete understanding of the distributional outcomes of international aid, scholars must consider both the direct effects of aid itself and its indirect effects on how local households and governments allocate resources they control. Incorporating insights from both the literature on comparative democratic participation (Verba et al. 1978; Berman 1997) and the literature on distributive politics (Franck and Rainer 2012; Golden and Min 2013), we examine differential crowding in and crowding out effects along political lines. We demonstrate that, when considering the full effects of international aid on distributive outcomes, there may be biases along political lines due to differential response of pre-existing institutions, even if the international aid itself is neutrally administered.

In analyzing the effects of international aid on distributional outcomes, we highlight the possibility that donor-funded NGOs may disappoint local citizens and governments, providing less benefit than expected. Aid via NGOs, and participatory development projects in particular, has rarely transformed socioeconomic outcomes (Casey et al. 2018; White, Menon, and Waddington 2018). Under these circumstances, we describe how co-partisans of the government who receive aid may become distributive losers.

We conducted a five-year randomized controlled study of a participatory development program in Ghana, and use this to study the complex interactions between participatory development aid and pre-existing political institutions and their combined effect on citizens' material well-being. Approximately half of 97 clusters of villages in Ghana's Eastern Region, each containing two villages, were assigned to partake in a multi-sectoral participatory development program run by The Hunger Project (THP), an international NGO with experience implementing similar programs in eight countries for more than a dozen years prior to the study. We tracked governance and socioeconomic outcomes using two waves of household, community and leadership surveys in these 194 villages. We collected long-term follow-up data (five years after baseline), as well as a breadth of information at the household, community and institution level. This allowed us to analyze how participatory development councils compared with and affected local traditional institutions and local governments, and how THP's participatory development program affected resource flows from other governance structures.

We contribute to two debates on the relationship between international aid and domestic politics in receiving countries. First, our study shows how participatory development institutions

can crowd out activities of pre-existing institutions. Existing studies have rarely explicitly focused on how new participatory institutions impact existing traditional institutions and local governments.⁶ Second, we highlight the complexity of international aid's distributional consequences. Debate typically has centered on how to avert biases toward government supporters, with empirical studies focusing on the distribution of aid inputs versus the distribution of socioeconomic outcomes (Brass 2012; Briggs 2012; 2014; 2017; Jablonski 2014). Bringing in theoretical insights from the literatures on comparative political participation and redistributive politics, we show how government-aligned citizens shifted resources into the participatory approach, and then ended up worse off because the new institutions performed poorly compared to pre-existing ones. By analyzing the broad socioeconomic effects of aid across sectors, as is considered best practice in the literature on distributional politics (Kramon and Posner 2013), we make an empirical advance in the study of international aid's distributional consequences.

II. Theorizing Participatory Development Aid's Cross-Institutional Effects and Distributional Consequences

Traditional government-to-government aid is subject to numerous problems, including elite capture and diversion for political purposes (de Mesquita and Smith 2009). Existing research has shown that donor-supported projects are frequently targeted at incumbent parties' core constituencies. In Kenya, Briggs (2014) shows that donor funds given to the government for

⁶ For important exceptions, see Beath, Christia, and Enikolopov (2013) and Deserranno, Nansamba, and Qian (2019)

specific projects were skewed to the incumbent president's base between 1989 and 1995; Jablonski (2014) demonstrates a similar pattern for government projects funded by the African Development Bank and the World Bank between 1992 and 2010. In Ghana, Briggs (2012) shows that a World Bank-funded electrification project was diverted to the incumbent's political base in the run-up to the 2000 elections. More generally, Hodler and Raschky (2014) shows that foreign aid is associated with higher levels of regional favoritism in countries with weak political institutions.

Donor support to NGOs, which has blossomed in the past two decades, is partly a response to these problems (Dietrich 2013). International aid to NGOs has been shown to be both less politically motivated than donor aid to governments (Büthe, Major, and Souza 2012; Faye and Niehaus 2012), and more neutrally allocated within countries; for example, Brass (2012) finds that support for the incumbent does not influence the location of NGO projects in Kenya.

Participatory development, or community development, approaches can be considered an extreme example of donor responses to government misallocation of aid. Participatory development aid is defined by its investment in new institutions that mobilize community members to participate in decision-making and project management (Mansuri and Rao 2013, 16). The exact form this investment takes varies, but it usually involves constituting new decision-making bodies and providing leadership training to community members with the goal of enhancing participation of previously excluded groups and individuals. The justification of this investment is based on the implicit assumption that aid will be misallocated without it, either due to intentional diversion by leaders of pre-existing government institutions or due to

inefficiencies in top-down approaches (Oates 1972; Ostrom 1996; Bardhan and Mookherjee 2000).

How do the new institutions created through participatory development interact with pre-existing institutions? In addition to any direct effect of participatory development programs on community decision-making and resource availability, they could have indirect effects on the inclusivity, accountability and resource allocation of pre-existing government. We note that the communities in which participatory development approaches are implemented virtually always have some pre-existing governance structures. They may have elected representatives in a legislature that oversees the formal state apparatus. Even in settings with weak formal government structures, there are typically pre-existing community-level institutions, often with customary legitimacy, that organize some types of local public goods (Murtazashvili 2016; Ostrom 1990).

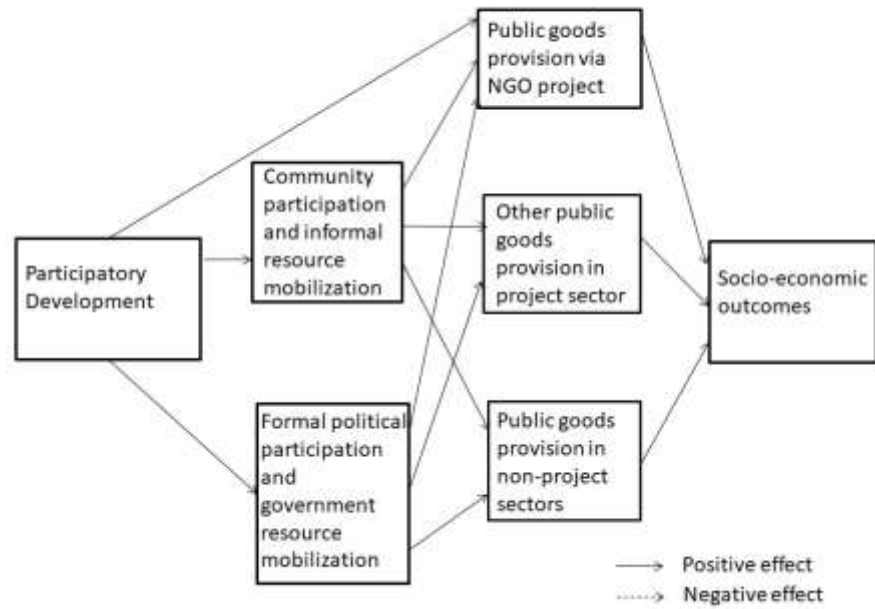
We start from the assumption that communities have both of these types of pre-existing political institutions and that individuals' well-being is a function of how responsive these institutions are to their needs. First, we consider the existence of informal community institutions that organize public goods and services using non-state resources. These include village traditional institutions, local business associations and sectarian organizations (Baldwin 2013; Cammett 2014; MacLean 2010), which may or may not have formal recognition from the government. The types of public goods organized by these institutions varies, but they typically depend on local voluntary contributions from citizens, often referred to as informal taxes, to finance the public goods they organize in the community (Lust and Rakner 2018; Olken and Singhal 2011). Second, we consider

the formal government, which can provide public goods and services in the community using state resources.⁷ In electoral democracies, citizens are typically represented in the formal government through an elected representative, who is responsible for advocating on their behalf.

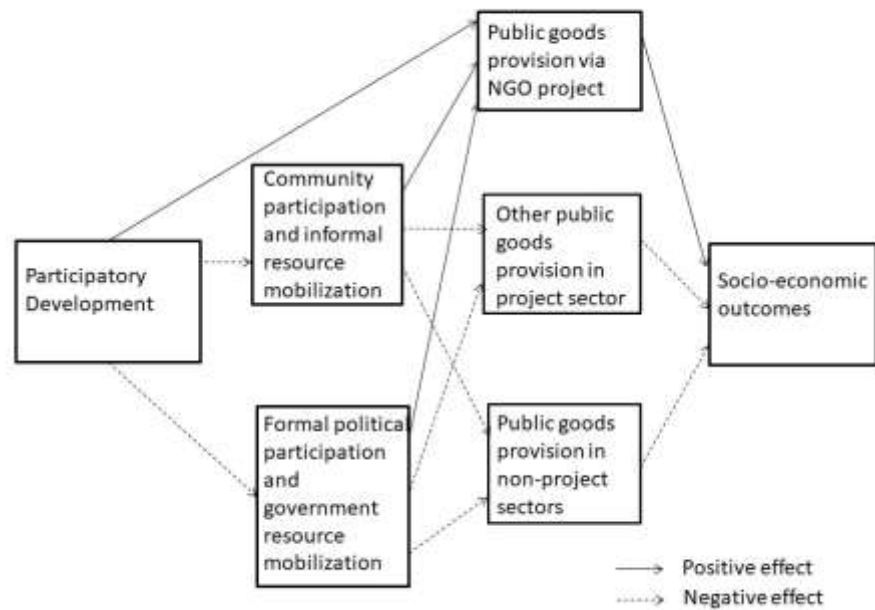
In this set up, a citizen's well-being is affected by NGO, household and government investment in public goods and services. Local households and governments must decide how much of their budgets to allocate to the NGO project, other projects in the NGO sector and other projects in the non-NGO sector within their community. The government may also decide to allocate resources to other communities. This system of institutions and investment decisions is depicted graphically in both panels of Figure 1, with each panel illustrating distinct types of effects participatory development aid might have on pre-existing political institutions.

⁷ We distinguish here between citizens' resources and state resources, given that an extremely small portion of state resources come from taxes on individuals in aid-receiving communities.

Figure 1. Cross-Institutional Effects of Participatory Development
(a) Mobilization Effects



(b) Displacement Effects



When models of participatory development have considered the potential interactions between participatory development institutions and pre-existing political institutions, they have tended to focus on the potentially positive *mobilization effects*. Specifically, they have emphasized that citizens gain participatory skills and coordination capacity that translates into their interactions with both informal community leaders and government representatives, thereby improving the responsiveness of these institutions and their investment in the community.⁸ For example, Casey, Glennerster and Miguel (2012) models community-driven development as providing participatory skills for new groups (through learning-by-doing and demonstration effects) and creating institutions such as community councils that reduce future coordination costs (in addition to subsidizing the cost of infrastructure materials). Avdeenko and Gilligan (2015) depicts participatory development as expanding social networks and thereby prosocial norms. Fearon, Humphreys and Weinstein (2015) emphasizes that these programs develop local leaders' coordination and mobilization skills. Implicit in these models is the assumption that there is significant slack in the performance of pre-existing political institutions, and so informal community institutions and governments could mobilize greater resources toward local public goods absent coordination or accountability failures. This model of interactions is depicted in Figure 1, panel A.

However, in cases in which there is not significant slack in the performance of pre-existing political institutions, participatory development projects may result in negative *displacement effects* in which citizens and governments reduce the amount of time and resources they channel

⁸ Even NGOs that do not devote significant resources to building new governance institutions have been argued to have these effects. See Boulding (2010; 2014).

through pre-existing institutions. Existing models of participatory development have paid insufficient attention to the possibility of these *displacement effects*. Citizens who are part of participatory development committees may not have the time or motivation to engage with community meetings or their formal government (Edwards and Hulme 1996; Nelson-Nuñez 2019). If skilled political leaders are scarce and participatory development projects are relatively ineffective at developing new leaders, participatory development projects could even lessen the quality of leadership in existing institutions through poaching effects (Moyo 2009; Deserranno, Nansamba, and Qian 2019).

Because household and government spending is fungible, citizens and governments may also reduce budget contributions they make to other projects in response to participatory development projects. Citizens contributing in cash or in kind to participatory development projects may pay less informal taxes to pre-existing community governing institutions (Lust and Rakner 2018; Olken and Singhal 2011).⁹ Governments with limited budgets may substitute resources away from sectors and communities served by participatory development programs (Khilji and Zampelli 1994; Torpey-Saboe 2015). These types of displacement effects could result in a reduction of public goods and service provision by traditional institutions and local governments, as depicted in Figure 1, panel B, potentially eliminating any distributional advantage of being an aid recipient.

⁹ Informal taxes differ from formal taxes in that they are not administered by the state or enforced by the formal legal system and therefore the amount paid is typically more elastic. In the extreme case of voluntary contributions, neutrality arguments suggest that the offset will be complete (Bergstrom, Blume, and Varian 1986).

Critically, both cross-institutional mobilization and displacement effects could plausibly differ along partisan lines. As a result, even if participatory development programs are provided equally to both supporters and opponents of the incumbent government, there may be partisan distributional consequences of participatory development. We draw on the existing literature on political mobilization and distributive politics to explain why this might be in the subsequent paragraphs, spelling out the distributional consequences of participatory development aid under various assumptions in Table 1.

Table 1. Distributional Consequences of Participatory Development Aid in Different Contexts

Type of effects of aid on pre-existing institutions	(2) If aid works	(3) If aid disappoints
1. Homogenous mobilization effects	Copartisans: recipients >> non-recipients Non-copartisans: recipients >> non-recipients	No distributional effects.
2. Mobilization effects for non-copartisans of government	Copartisans: recipients > non-recipients Non-copartisans: recipients >> non-recipients	No distributional effects.
3. Mobilization effects for co-partisans of government	Copartisans: recipients >> non-recipients Non-copartisans: recipients > non-recipients	No distributional effects.
4. Homogenous displacement effects	No distributional effects.	Copartisans: recipients < non-recipients Non-copartisans: recipients < non-recipients
5. Displacement effects with co-partisan government spending	Copartisans: recipients = non-recipients Non-copartisans: recipients > non-recipients	Copartisans: recipients < non-recipients Non-copartisans: recipients = non-recipients

In the first column of Table 1, we outline the distributional consequences of participatory development aid, assuming that aid generates benefits for at least some sub-groups in the population. In the first row, we consider how cross-institutional mobilization effects alter the effects of participatory development. The expectation is that mobilization effects ratchet up the

benefits of participatory development for aid recipients, as they result in increased investment in public goods and services from pre-existing governance institutions as well as the NGO.

In the second and third rows, we consider the possibility of partisan bias in the size of mobilization effects. In theorizing the direction of partisan effects, we start from the observation that individuals affiliated with the national incumbent party typically believe their governments are more democratic and more responsive to them (Anderson and Tverdova 2003; Norris 2011). However, the existing literature provides two competing predictions about whether this should cause greater mobilization effects for incumbent co-partisans or other citizens, depending on whether scholars view civil society interventions as having the ability to substitute for party organizations that connect citizens to government or as necessarily being complemented by partisan mobilization.

One strand of the literature emphasizes the importance of citizens' civic skills, attitudes and associational memberships in explaining democratic political participation (Almond and Verba 1963; Putnam, Leonardi, and Nanetti 1993). From this theoretical perspective, the training and coordination of citizens who are unaligned with the incumbent would be expected to generate particularly large effects on their engagement with pre-existing governance institutions, given their lower baseline levels of political efficacy. In this case, we would expect especially large benefits to being an aid recipient for individuals who are not co-partisans of the incumbent government, as outlined in row 2.

In contrast, a second strand of the literature on political participation emphasizes the unique role of political parties in mobilizing democratic engagement (Verba et al. 1978; Berman 1997). From

this perspective, increases in individuals' civic skills, attitudes and organizational capacity are only likely to increase democratic political participation if they are also connected to political parties that allow their opinions to be articulated in the centers of government power. Insofar as opposition political parties have weak abilities to do this in new democracies, this theoretical perspective leads to a competing prediction: we would expect citizens who are aligned with the incumbent party to experience the largest mobilization effects. In this case, we would expect especially large benefits to being an aid recipient for individuals who are co-partisans of the incumbent government, as outlined in row 3.

In the bottom two rows of the table, we consider the distributional consequences of displacement effects. Displacement effects could result in a reduction of public goods and service provision by community-level institutions and local governments, potentially eliminating any distributional advantage of being an aid recipient, as indicated in row 4. However, this may also be subject to partisan biases. Specifically, we are interested in whether co-partisans of the government are likely to be subject to larger or smaller displacements of government spending as a result of participatory development projects.

Here, the existing literature provides clear expectations. Individuals who are co-partisans of the government are typically favored in their access government-financed local public goods and services (Burgess et al. 2015; Franck and Rainer 2012; Golden and Min 2013; Kramon and Posner 2016).¹⁰ As a result, on average, the absolute value of state resources displaced as a result of

¹⁰ In empirical studies of sub-Saharan Africa, core support is often measured through co-ethnicity instead of co-partisanship, given the strong correlation between the two in some countries. We have measures of both co-ethnicity and co-partisanship but we prefer to focus on the latter given

participatory development programming should be higher for government co-partisans. In this context, we would expect the receipt of aid to make a larger difference to the well-being of individuals who are not government co-partisans. In the extreme case in which the government only invests resources to benefit co-partisans (and the government has at least as many resources as the NGO), aid recipients are only better off than non-aid recipients if they are not copartisans, as in row 5.

In our discussion of the distributional consequences of participatory development aid so far, summarized in the first column of Table 1, we have described aid as having a positive impact. However, many international aid projects, including participatory development ones, have limited impacts (Casey et al. 2018; King and Samii 2014; White, Menon, and Waddington 2018; Wright and Winters 2010), which might not be understood by government policymakers and households in making their own resource allocation decisions. Casey, Glennerster, Miguel and Voors (2018) surveys the expectations of academics, international policymakers and domestic policymakers on the long-term effects of a community-driven development project they had implemented in Sierra Leone between 2005 and 2009; despite the fact that they had demonstrated no medium-term institutional effects of the program in Sierra Leone in a published paper (Casey, Glennerster, and Miguel 2012), policymakers in the country expected large positive effects. More generally, public opinion data from sub-Saharan Africa suggests that many citizens have extremely high expectations for NGOs. According to data from Afrobarometer survey (4th round), 43 percent of Africans (across all sampled countries) believe that international donors

that partisanship is a stable identity in Ghana and multiple ethnic groups consistently support each of the two major parties.

and NGOs help their country a lot; an additional 33 percent say they help somewhat and only 5 percent say that they do not help at all.

As a result, in the third column of Table 1, we consider the distributional consequences of aid that has a disappointing impact. In contexts without displacement effects (rows 1-3), there are limited distributional consequences of participatory development aid projects that have limited effects. However, once we introduce the possibility of displacement effects, aid that fails to meet citizens and governments' expectations has distributional consequences. Aid recipients may reallocate too much effort to aid projects, given their inflated expectations regarding their benefits. Similarly, governments may displace too many resources from aid recipients. If there is no slack in citizens' and governments' budget constraints and aid ultimately has disappointing effects, aid recipients are expected to be worse off than non-aid recipients in the short to medium-term, as indicated in row 4.

An additional wrinkle is introduced if the government is biased in favor of its copartisans in its displacement of resources from aid projects that ultimately underperform. Row 5 considers the extreme case in which the government only spends state resources on its copartisans, showing that copartisan recipients of aid will be worse off than copartisans who do not receive aid in this context, while there will be no difference in well-being among non-copartisans. In a world in which aid disappoints and the government is biased in favor of its copartisans, copartisan aid recipients can become distributive losers.

Thus, our ambition is twofold. We analyze whether participatory development aid has mobilization and/or displacement effects on pre-existing political institutions, as depicted in Figure 1. We also analyze whether these effects differ along partisan lines, as described in Table 1.

III. Local Governance in Ghana's Eastern Region

We study participatory development's effects on preexisting governance structures in the context of Ghana's Eastern Region. Community-driven development projects are common across low and middle income countries (Mansuri and Rao 2013; White, Menon, and Waddington 2018). However, most existing experimental research on the effects of community-driven development has focused on post-conflict settings and "failed states" (Avdeenko and Gilligan 2015; Casey, Glennerster, and Miguel 2012; Fearon, Humphreys, and Weinstein 2015; Humphreys, Sierra, and Windt 2014). We study the impact of participatory approaches in a poor but peaceful setting with strong pre-existing political institutions. The effects of participatory development programs on preexisting institutions are arguably particularly important in settings with strong existing governance structures. Our study took place in villages across Ghana's Eastern Region.¹¹ These communities are governed by traditional chieftaincy institutions, in addition to elected local governments and national governments.

¹¹ Only four of the 17 districts in existence in 2006 were excluded – one because it was urban; two because the program had previously been rolled out in these districts; and one because we were not able to successfully collect baseline data in this district.

The chieftaincy structure is broadly similar across our study communities.¹² At the top of the traditional hierarchy is the chief (*omahene*), with divisional chiefs (*ohene*) and village chiefs (*odikro*) below them. For most rural citizens, the most relevant of these leaders are village chiefs, who are selected from within the village's ruling family and typically rule for life. They normally govern their villages with the assistance of a council, which includes other family heads (*abusuapanyin*) and elders (*panyin*) (Arhin 1985). Village chiefs play critical roles in local dispute resolution, land allocation, meeting organization and community mobilization. However, they do this without salaries, budgets, or formal support from the government.¹³ Instead, they depend on informal norms to underpin their power and voluntary contributions from community members to accomplish projects ("self-help projects").

In parallel to the chieftaincy structure, communities in Eastern Ghana are also governed by district governments. Much of the power lies with the District Chief Executive, who is appointed by the president and combines executive and administrative functions. As a result, the party winning the national presidency has significant control over the allocation of resources within districts. Each district also has a district assembly; two thirds of its members are popularly elected from single-member electoral districts composed of groups of villages/neighborhoods (with total populations of around 10,000 each) and the other third is appointed. District elections are held every four years, with one set held during our study (in late 2010/early 2011). Officially, these

¹² This is despite the fact that the region is made up of multiple (Akan and Krobo) ethnic groups, reflecting the historical influence of Akan practices on their neighbors and colonial-era standardization. See Wilson (1987).

¹³ Higher level traditional leaders, such as chiefs, do have official roles and receive some resources from the Ghanaian government.

elections are non-partisan, although the political affiliations of candidates are often well-known locally, and the position of assemblymember is a part-time volunteer position. The district assembly is responsible for approving the district budget and providing oversight of the district administration. District assemblymembers are expected to lobby for resources from the district budget to support local projects, especially in the areas of basic education, primary health care, local roads, environmental protection, water and sanitation (Crook et al. 1998). In all of our study areas, the vast majority of the district budget comes from transfers from the national government using a formula-based fund.

Citizens also participate in national elections to determine control of the parliament and presidency. National politicians are inaccessible to most rural Ghanians but these elections structure partisan identities, and are deeply competitive between two major parties, the National Democratic Congress (NDC) and the New Patriotic Party (NPP). The two parties have strong regional and ethnic bases of support, and many Ghanaians have stable partisan preferences. For example, Lindberg and Morrison (2005) finds that 82 percent of parliamentary voters in the 2000 election had voted for the same party in 1996, and Weghorst and Lindberg (2013) finds that only 22 percent of voters split their presidential and parliamentary vote between different parties in any of the three elections covered by their study (1996, 2000 and 2004). The NDC was the national incumbent party for almost all of the period of our study, taking over the presidency after the December 2008 election, and winning re-election in December 2012.

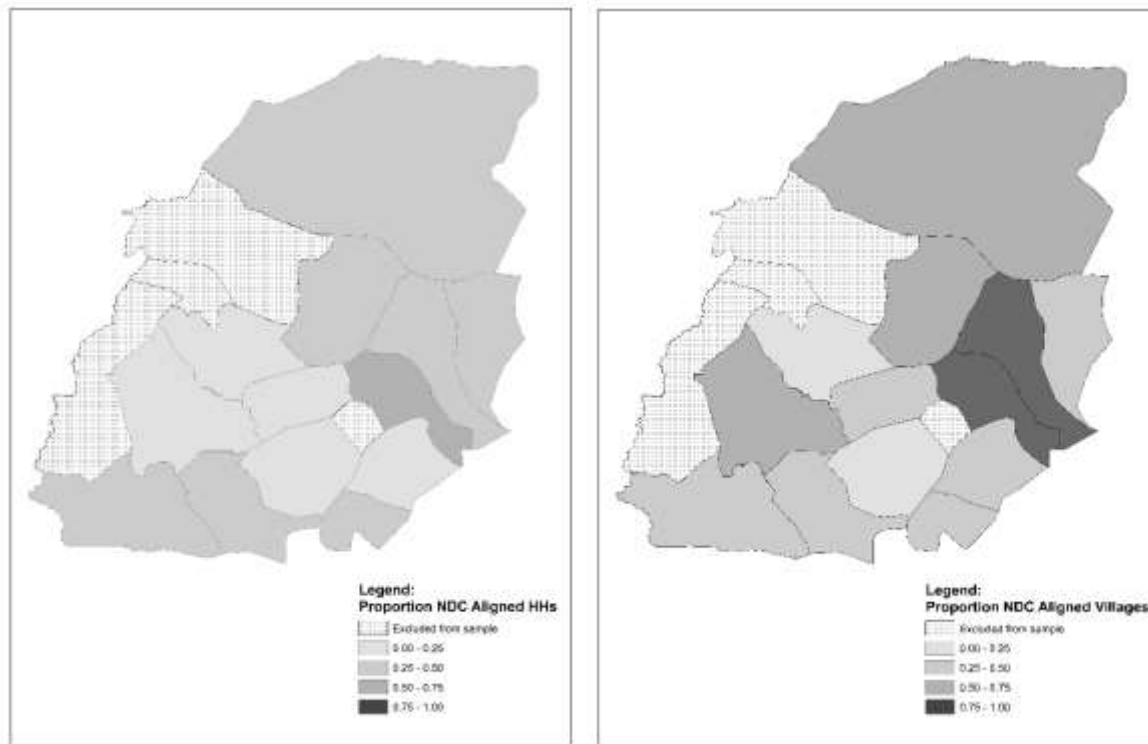
Ghana's Eastern Region is uniquely divided between NDC and NPP supporters, largely due to the fact it includes both Ewe and Krobo ethnic groups (traditionally support the NDC) as well as Akan

groups (Akyem and Akuapem, traditionally support the NPP).¹⁴ Importantly, NPP and NDC supporters are intermingled within districts and even villages in our study, with 95 percent of villages containing households that supported different parties at baseline. Figure 2 displays the distribution of NDC support across the region at baseline, indicating the proportion of NDC-aligned households within our study villages in each district in the image on the left and the proportion of NDC-aligned villages (defined as villages in which at least 30 % of households are NDC aligned) in the image on the right. Copartisanship with the national government is extremely important for distributive outcomes in Ghana, influencing the distribution of funds from both the national government and district governments, given the role of the president in appointing the District Chief Executive (Asunka 2017; Nathan 2019).

Thus, prior to the expansion of participatory development institutions in the region, the study villages already had hereditary chiefs who governed them at the village level, and elected leaders who represented them within District Assemblies. Participatory development aid could plausibly have positive mobilization spillovers and/or negative displacement effects on the responsiveness of each of these institutions to citizens. Furthermore, given the strength of partisan affiliations and the history of redistribution along partisan lines in Ghana, these effects could plausibly differ depending on whether citizens are co-partisans of the national government.

¹⁴ In a regression model predicting NDC affiliation at the household level in our sample, only ethnic variables and the percentage of women in the household are statistically significant at conventional confidence levels. See Appendix A.

Figure 2. NDC Co-Partisanship Across Study Districts



IV. Intervention and Experimental Research Design

Our analysis of the distributional consequences of participatory aid is built around a randomized controlled trial of The Hunger Project's (THP's) activities in Eastern Ghana. THP is a major international NGO whose approach seeks to empower men and women to take control of their futures by mobilizing them to act collectively within their local communities. In particular, THP seeks to cultivate stronger leadership within communities both by organizing workshops that train participants in leadership skills and by creating new inclusive governance structures.

The broad components of THP's approach (described in Appendix B) exemplify the participatory development approach that has become prevalent in the aid industry. Community members are

involved in project oversight in part to help align projects with community needs, but also to provide on-the-ground monitoring and reduce dependence on outside resources in the context of project implementation. In the THP model, as in many recent community-driven development programs, a great deal of focus is building the capacity of communities to work together to overcome socioeconomic challenges outside the narrow context of administering program funds. Community members are expected to devote significant resources in cash or in kind to supplement the donor funds provided for programming activities, and the goal is to have the local government provide support for many of the programs subsequently run out of the center.

The THP approach is also explicitly multi-sectoral. The THP provides financial support for a variety of programming activities, which are run out of community centers it helps local communities to construct. These centers contain meeting halls, clinics, rural banks, foodbanks, toilets, a demonstration farm, and either a preschool or library, and THP also supports agricultural training programs, adult literacy classes and microfinance programs.

Our study took place in 194 villages, divided into 97 two-village groupings, across 13 districts in Eastern Region. The village groupings were randomly assigned to treatment (57) and control (46) through district-level lotteries, as described in Appendix C. Not all of the village groupings invited to take part in THP's programming accepted the invitation. Following these workshops, just over half of the villages (in 28 of 51 treatment groupings) actually began the THP process. All but three of these groupings successfully completed construction of a community center, and four groupings built two community centers. In Appendix D, we show that randomization yielded statistically similar groups (i.e., we fail to reject the null that treatment assignment is orthogonal

to the baseline attributes of our study communities), as well as the differences between the communities within the treatment group that took-up as compared to those that did not.

THP approximated the ideals of the participatory development approach in important ways. First, it successfully created new participatory development institutions with more diverse leaders than existing hereditary and elected institutions, as we show in Appendix E. Second, it was successful in exposing a large proportion of adult community members to its activities, and exposure was not biased along partisan lines, also demonstrated in Appendix E. As a result of these successes in implementation, the program arguably represents a best case for considering whether participatory development can have positive cross-institutional effects on engagement with preexisting political institutions.

We are able to assess the effects of participatory development approaches on participation in, leadership of and investment through various governance institutions by bringing together four types of data, collected at multiple points in time. The timing of the distinct data collection efforts relative to programing activities are displayed in Figure 3 and described below:

Household surveys. In each of the 97 village groupings in the study, two villages were randomly selected for surveying. A baseline survey was conducted in 2008, at which point none of the study villages had built the community center that is the centerpiece of THP's programming. Twenty households were randomly selected for interviewing in each village in the sample, except in the handful of cases where the village contained fewer than 20 households. A follow-up survey was conducted with the same households in 2013. At this point, all of the treatment villages had been introduced to THP's programming at least two years earlier, and some had been introduced to it

five years earlier, as illustrated in Figure 3. Given the long timeframe of the study, attrition was a significant risk. We were able to resurvey 74 percent of baseline households. We have examined whether the treatment – either by itself or in interaction with baseline outcome variables – affects the likelihood of attrition, and have found no evidence that suggests concerns of bias due to attrition from the survey sample frame, as demonstrated in Appendix F.

Community leader surveys. We surveyed a key informant from each village (most frequently, the village chief or another local traditional leader) about local services at baseline and as part of our follow-up surveys. In our follow-up surveys, we also surveyed the area’s representative in the district government (the district assemblyperson).

Administrative data on local election returns and candidates. We obtained the official local election returns and candidate forms for the local government elections held in the end of 2010 and the beginning of 2011 from the Electoral Commission of Ghana. We consider only the electoral areas containing study villages in our analysis (N=122). Many electoral areas contain two study villages from the same village grouping; only three contain villages from different village groupings assigned to both treatment and control. We code electoral areas as treated if they contain any study villages assigned to treatment. By the time of these elections, the vast majority of the treated communities had been exposed to THP’s programming, as Figure 3 illustrates, although many had had a completed community center for less than a year.

The statistical analysis of the effects of the NGO’s programming is complemented with evidence from a qualitative follow-up study conducted in 12 communities in 2015, the method and results of which are described in Appendix G.

Figure 3. Timeline of data collection and program roll out



V. Results: Cross-Institutional Effects and Distributional Consequences

We are interested in whether participatory development aid has cross-institutional mobilization effects that improve engagement with pre-existing hereditary and elected leadership and/or displacement effects that reduce investment through pre-existing institutions. In addition, we consider whether there are partisan biases in either of these effects. We begin by considering the evidence for each type of cross-institutional effect, using household, leadership and administrative data on participation, accountability and investment in local public goods, before considering the aggregate effects of participatory development on the distribution of socioeconomic outcomes. Due to the imperfect take-up of the programming among treated communities, we estimate both the “intent to treat” (ITT) and “treatment on the treated” (TOT)

effects, using assignment to treatment as an instrument for mobilizing to receive programming at the village level in the latter case.

We evaluate effects by constructing indices for each area of hypothesized impact. This provides a clearer picture of the overall effect of the participatory approach in each area, and helps address the problem of multiple hypothesis testing. Each index is created from a group of variables measuring outcomes associated with the concept of interest by averaging the standardized sub-components, and then re-standardizing the index.¹⁵ As a result, the effect of the program on the indices should be interpreted in terms of standard deviations of the index within the control group.

We examine the effects of participatory programming at two different levels of analysis, depending on the unit of measurement. Many of our measures come from our household survey, in which case outcomes are measured at the household level. In addition, we have measures of local government investment and measures of political participation in local government measured at the level of the electoral district (called “electoral areas”).

The Intent to Treat (ITT) estimate of the effect of THP on household-level outcomes is $\hat{\beta}_1$ from the following OLS regression specification:

$$y_i = \beta_0 + \beta_1 THP_j + \beta_2 X_i + D_k + \epsilon_i$$

¹⁵ In some cases, the sub-components are also themselves indices of variables, as explained in Appendix H.

where i indexes households, j indexes village groupings, and k indexes districts. THP_j is an indicator variable that takes a value of 1 if the village grouping was assigned to treatment in the lottery, X_i is the baseline measure of the outcome variable (where available), and D_k are district fixed effects. In cases where baseline data was available for some but not all observations, we dealt with missing data using dummy variable adjustment. The error term is clustered at the village grouping level. For electoral area level outcomes, we replace y_i with y_{EA} , and THP_j is a variable that takes a value of 1 if any sampled village in the electoral area was assigned to treatment and 0 if all sampled villages in the electoral area were assigned to control; in these models, standard errors are clustered by village groupings.¹⁶ Given imperfect take-up at the village-level, we also estimate the Treatment on the Treated (TOT) using an instrumental variable estimator implemented using the Generalized Method of Moments (GMM).¹⁷

In addition to estimating the models on the entire sample, we also estimate separate models by baseline support for the party of the president during the major period of the study, the NDC (see Figure 3). For the household-level analysis, we estimate separate models for households in which the majority of respondents identified as NDC supporters in our baseline survey, which was conducted just prior to the 2008 national elections (28 percent of households), and households in which the majority of respondents did not identify as NDC supporters, either because they supported other parties or had no political allegiance (72 percent of households).

¹⁶ In cases where villages in the same electoral area fall in different village groupings, we have joined the two village groupings for the purpose of calculating standard errors.

¹⁷ The first stage results are included in Appendix I.

In the village-level analysis, we distinguish between villages in which at least 30 percent of households are affiliated with the NDC and those without.¹⁸

First, we use household survey data in Table 2 to examine whether there are effects on citizen's participation in village-level governance, their perceptions of the accountability of the village chief, and their perceptions of the accountability of their district assembly member. Our measure of participation in village governance is an index averaging associational membership, village assembly attendance and village assembly contributions. Our measure of the accountability of the village chief's leadership is an index averaging the village chiefs' accessibility, openness to dissent and trustworthiness. Our measure of the accountability of the assemblymember is an index averaging assemblymembers' accessibility, perceived responsiveness and trustworthiness.

We find that participatory development increased participation in village-level governance for members of the NDC only. Focusing on the effect of the treatment on the treated (TOT) across the entire sample, we observe an increase in participation of 0.10 standard deviations ($se=0.08$), which is not statistically significant at conventional levels. However, there are heterogeneous effects depending on partisan affiliation. For NDC-affiliated households, the effect is 0.40 standard deviations ($se=0.17$), which is statistically significant at the 95 percent confidence level; for all other households, we estimate a small and not statistically significant negative effect (effect = -0.09 standard deviations; $se=0.12$).

¹⁸ This cut-off was chosen because it represents an above-average level of support for the NDC in rural Eastern region, where just over 28 percent of our respondents felt an affiliation toward the NDC. In Appendix J, we show the results are robust to different cut-offs.

We find more consistently positive effects on perceptions of the quality of the village chiefs' leadership. Using the TOT estimates, we find a positive effect of 0.21 standard deviations across the entire sample ($se=0.09$), which is statistically significant at the 95 percent confidence level. The estimated effect size is larger and more statistically significant for NDC-affiliated households, but the effects are positive regardless of partisan affiliation. In contrast, we find no evidence that participatory development changed citizens' perceptions of the accountability of the district assembly members, either across the sample as a whole or in either partisan subgroup.

Next, we look for evidence of mobilization effects in community-level data in Table 3, with the outcomes collected from electoral data measuring the participation of voters and candidates in the 2010/2011 district elections and leadership survey data measuring the participation of local assembly members in district government. Focusing on the TOT estimates, voter turnout *decreased* on average by 10 percentage points ($se = 4pp$) in communities that took up the treatment, an effect that is statistically significant at the 95 percent confidence level. However, these negative effects are concentrated entirely within villages with below average levels of NDC support, where voter turnout decreased on average by 17 percentage points ($se=8pp$); among NDC-affiliated villages, participatory development had a small and not statistically significant effect on voter turnout (effect = -4 percentage points, $se=6pp$).

In contrast, there appear to have been positive mobilization effects at the candidate level. Focusing on the TOT estimates, we find an average increase of 0.52 candidates running for office in the 2010/2011 local government elections ($se=0.30$), which is statistically significant at the 90 percent confidence level. However, these effects are concentrated entirely within NDC-affiliated

villages, where we find an average increase of almost 2 additional candidates running for office ($se=0.51$), which is statistically significant at the 99 percent confidence level. In contrast, in villages with below average support for the NDC, participatory development is estimated to have a slightly negative but not statistically significant effect on the number of candidates for office.

Finally, we consider how active the assemblymember elected in the 2010/2011 local government elections reported being in office. We use data from our interviews with assembly members to create an index of their activity level, averaging the district assembly members' attendance at district assembly meetings, the number of times they raised issues in district assembly meetings, the number of times they met one-on-one with their DCE, the number of times they met with community leaders, the number of times they met with voters, the number of infrastructure projects they facilitated and the number of NGOs (excluding THP) whose activities they facilitated.

Table 2. Village Participation and Local Accountability (Household Survey Data)

		Entire Sample				NDC Aligned HHs				Non-NDC Aligned HHs				
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	
		ITT	TOT	Control	N	ITT Effect	TOT	Effect	Control	ITT	Effect	TOT	Control	N
		Effect	Effect	mean		(st.	(st.		mean	(st. error)	(st. error)	Effect	mean	
		(st. error)	(st. error)	(st. dev.)		error)	error)		(st. dev.)			(st. error)	(st. dev.)	
Community		0.054	0.103	0.000	2746	0.214*	0.400*	-0.026	680	-0.042	-0.089	0.039	1704	
Participation		(0.045)	(0.082)	(1.000)		(0.096)	(0.172)	(0.960)		(0.052)	(0.115)	(1.022)		
Index														
Village	Chief	0.111*	0.211*	0.000	2744	0.175*	0.324*	0.114	680	0.099+	0.211+	-0.075	1703	
Accountability		(0.047)	(0.091)	(1.000)		(0.072)	(0.131)	(1.004)		(0.057)	(0.122)	(0.988)		
Index														
District		0.069	0.131	0.000	2743	-0.050	-0.092	0.110	680	0.046	0.098	-0.007	1702	
Assemblymember		(0.072)	(0.131)	(1.000)		(0.096)	(0.179)	(0.969)		(0.076)	(0.157)	(1.020)		
Accountability														
Index														

Notes: +significant at 10%; * significant at 5%; ** significant at 1%. Column (1) presents OLS estimates (with standard errors reported in parentheses), clustered at the unit of randomization (village cluster), and controlled for district effects. Each row reports results for a single OLS regression. Column (2) reports IV-GMM treatment-on-the-treated estimates (with standard errors reported in parentheses) with mobilizing to receive an epicenter being the first stage clustered at the unit of randomization (village cluster). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations. Columns 5-8 report the same entities using the sample of NDC-aligned households. Columns 9-12 report the same entities using the sample of non-NDC aligned households.

Table 3. Local Government Participation and Representation (Electoral Data and Leadership Surveys at Electoral District Level)

		Entire Sample				NDC Aligned Villages (>=30 % NDC HHs)				Non-NDC Aligned Villages (< 30 % NDC HHs)			
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
		ITT Effect (st. error)	TOT Effect (st. error)	Control mean (st. dev.)	N	ITT Effect (st. error)	TOT Effect (st. error)	Control mean (st. dev.)	N	ITT Effect (st. error)	TOT Effect (st. error)	Control mean (st. dev.)	N
Voter turnout in district elections (proportion)		-0.051* (0.025)	-0.095* (0.042)	0.502 (0.143)	111	-0.019 (0.040)	-0.037 (0.062)	0.471 (0.122)	44	-0.086* (0.040)	-0.174* (0.078)	0.502 (0.143)	51
Number of candidates		0.278+ (0.167)	0.523+ (0.299)	2.526 (0.804)	122	0.975** (0.233)	1.905** (0.505)	2.143 (0.727)	49	-0.155 (0.239)	-0.291 (0.486)	2.800 (0.761)	62
District Assemblymember Activity Index		0.419+ (0.225)	0.759+ (0.396)	0.000 (1.000)	106	0.765* (0.333)	1.802* (0.759)	-0.350 (1.053)	46	0.004 (0.272)	-0.130 (0.463)	0.296 (0.880)	53

Notes: + significant at 10%; * significant at 5%; ** significant at 1%. Column (1) presents OLS estimates (with standard errors reported in parentheses), clustered at the unit of randomization (village cluster), and controlled for district effects. Each row reports results for a single OLS regression. Column (2) reports IV-GMM treatment-on-the-treated estimates (with standard errors reported in parentheses) with mobilizing to receive an epicenter being the first stage clustered at the unit of randomization (village cluster). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations and the unit of observation. Columns 5-8 report the same entities on the sample of villages with higher than average baseline support for the NDC. Columns 9-12 report the same entities using the sample of villages with lower than average baseline support for the NDC.

Across the entire sample, we find a positive mobilization effect on district assembly members' activities. Focusing on the TOT estimates, THP increased elected representatives' reported activity levels by 0.76 standard deviations ($se=0.41$), a substantively large effect that is statistically significant at the 90 percent confidence level. However, the effect is concentrated entirely within villages with high support for the NDC, where the increase was 1.8 standard deviations ($se=0.76$), which is statistically significant at the 95 percent confidence level. In contrast, the estimated effect on participatory development in villages with low levels of NDC support is very small, though estimated with considerable error (effect $=-0.13$ standard deviations; $se=0.46$).

Taken together, the evidence in Tables 2 and 3 suggests that participatory development has cross-institutional mobilization effects in this context, but only for those who were politically aligned with the incumbent government. In households and villages affiliated with the NDC, we find positive and significant mobilization effects for 4 of the 6 outcomes considered. In contrast, for households and villages that do not strongly support the NDC, we do not see a consistent pattern in the effects, and we even observe a significant negative effect on voter turnout in the district elections. Importantly, these differences in mobilization effects are not a result of different exposure to THP. Appendix Table D1 shows that we do not observe partisan differences in participation within treatment villages. Instead, it appears that the skills and capacity developed through THP need to be complemented with partisan connections to the centers of government power in order to translate into increased levels of engagement.

Next we consider the effects of participatory development on investment in local public goods through preexisting institutions. On the one hand, the observed improvements in engagement with these institutions could plausibly result in greater investment, resulting in a positive effect. On the other hand, these institutions may be less willing or able to funnel resources into local public goods once these are being provided through participatory development institutions, causing a negative displacement effect. We consider the effects of participatory development on two streams of investment in local public goods – voluntary contributions from households to fund projects and district government investment in local projects. The first type of investment is often mobilized through traditional village institutions, while the second type of investment is the result of district-level representation and investment decisions.

In Table 4 Panel A, we consider the effect of participatory development programming on household contributions to self-help projects other than the epicenter. We calculate the value of each household's contributions to public goods as the sum of their monetary and labor contributions to local public goods other than the epicenter in the previous twelve months.¹⁹

We find that participatory development programming decreases voluntary contributions to other local projects. The TOT effect is a 9.7 GHS decrease ($se=5.7$) in the value of contributions, which is statistically significant at the 90 percent confidence level. However, the decrease in voluntary contributions appears to be concentrated more within NDC-aligned households; here we observe a 26.7 GHS decrease ($se=15.9$), which is statistically significant at the 90 percent confidence level.

¹⁹ We impute the value of labor contributions by multiplying the number of (eight hour) days worked by the typical daily wage for an unskilled agricultural task (weeding) in the village; data on the typical daily wage for men and women was collected as part of our community survey.

Among non-NDC aligned households, we observe a smaller 7.2 GHS decrease ($se=6.05$), which is estimated with considerable error. If we distinguish between voluntary contributions to projects in sectors in which THP explicitly works (health, water, micro-finance, sanitation and community center construction) and projects in sectors in which THP does not work, we observe a larger decrease in contributions to projects in sectors in which THP is working across the sample as a whole and also in the sample of NDC households, but the point estimate on contributions to public goods in other sectors is also negative (though measured with a large amount of error).

In Table 4 Panel B, we consider the effect of THP programming on the scope of projects financed by the local government in the electoral area in the most recent electoral term (2011-2014). As part of our community survey, we collected information on whether the local government financed projects in nine different sectors during this time period -- health, water, sanitation, childcare, micro-finance, education, road, power and agricultural processing. We measure local government investment as the proportion of these sectors in which they financed a project between 2011 and 2013.²⁰

We estimate no change in the proportion of sectors in which the local government financed projects across the sample as a whole. Interestingly, despite the fact that NDC-aligned villages experienced larger increases in political participation as a result of participatory development, there is little evidence that they managed to increase government investment through this engagement; in fact, there is a 9.2. percentage point decrease in local government investment

²⁰ Unfortunately, we were unable to collect reliable data on the amount invested in each project.

associated with participatory development in NDC-aligned villages, but the estimate is measured with considerable error ($se=0.083pp$).

The effect on overall government investment hides differences between government investment in sectors on which THP efforts were concentrated and sectors in which THP placed less emphasis. Focusing on the TOT effect, we see a reduction of 6.8 percentage points ($se=0.04pp$) in the proportion of THP sectors with local-government financed projects, essentially eliminating any government investment in these sectors. In contrast, we find an increase of 7.4 percentage points ($se=0.04pp$) in the proportion of non-THP sectors with local-government financed projects. Both of these effects are statistically significant at the 90 percent confidence level. When we split the sample between NDC-aligned and non-NDC-aligned villages, the effects on investment in different sectors are each estimated with considerable error, but with suggestive evidence that the increase in non-THP sectors is concentrated in non-NDC-aligned villages.

Thus, Table 4 indicates that any positive effects of participatory development on engagement with pre-existing institutions did not result in greater investment in local public goods through these institutions. For the NDC-aligned households who experienced the largest improvements in political engagement as a result of the program, we observe negative displacement effects in citizens' contributions to other local public goods (statistically significant at the 90 percent confidence level) and suggestive evidence that local governments might have displaced funds from these communities too. Although participatory development may have improved engagement with pre-existing institutions on some dimensions, this was not associated with increased ability to mobilize resources behind community-level projects.

Table 4. Mobilization of Public Goods by Households (Panel A) and Government (Panel B)

PANEL A: HH MOBILIZATION	ENTIRE SAMPLE				NDC Aligned HHs				Non-NDC Aligned HHs			
	(1) ITT Effect (st. error)	(2) TOT Effect (st. error)	(3) Control mean (st. dev.)	(4) N	(5) ITT Effect (st. error)	(6) TOT Effect (st. error)	(7) Control mean (st. dev.)	(8) N	(9) ITT Effect (st. error)	(10) TOT Effect (st. error)	(11) Control mean (st. dev.)	(12) N
HH contributions to non-THP public goods (cedis)	-5.10+ (2.90)	-9.73+ (5.71)	15.31 (84.00)	2745	-14.33+ (8.087)	-26.72+ (15.92)	22.45 (139.98)	680	-3.38 (2.79)	-7.22 (6.05)	14.08 (61.5)	1704
HH contributions to public goods in THP sectors (cedis)	-3.73 (2.41)	-7.10 (4.66)	4.24 (67.31)	2745	-10.66 (8.22)	-19.88 (15.70)	11.18 (137.42)	680	-1.26+ (0.72)	-2.69+ (1.56)	2.24 (15.0)	1704
HH contributions to public goods in non-THP sectors (cedis)	-1.38 (1.97)	-2.63 (3.78)	11.08 (50.45)	2745	-3.67 (2.41)	-6.84 (4.66)	11.27 (28.68)	680	-2.12 (2.71)	-4.52 (5.81)	11.84 (59.73)	1704
PANEL B: GOVT MOBILIZATION	ENTIRE SAMPLE				NDC Aligned Villages (>=30 % NDC HHs)				Non-NDC Aligned Villages (< 30 % NDC HHs)			
Proportion of sectors with local gov funded projects	0.005 (0.033)	0.006 (0.053)	0.072 (0.162)	117	-0.056 (0.054)	-0.092 (0.083)	0.092 (0.223)	48	0.035 (0.052)	0.074 (0.082)	0.065 (0.126)	58
Proportion of THP sectors with local gov funded projects	-0.038+ (0.022)	-0.068+ (0.036)	0.054 (0.158)	116	-0.063 (0.054)	-0.112 (0.083)	0.070 (0.223)	48	-0.038 (0.023)	-0.060 (0.041)	0.049 (0.118)	57
Proportion of non-THP sectors with local gov funded projects	0.044+ (0.024)	0.074+ (0.040)	0.025 (0.048)	115	0.006 (0.016)	0.020 (0.027)	0.025 (0.050)	47	0.071 (0.046)	0.128+ (0.074)	0.026 (0.049)	57

Notes: +significant at 10%; * significant at 5%; ** significant at 1%. Column (1) presents OLS estimates controlling for district effects (with standard errors, reported in parentheses, clustered at the unit of randomization, the village cluster). Each row reports results for a single OLS regression. Column (2) reports IV-GMM treatment-on-the-treated estimates with mobilizing to receive an epicenter instrumented by treatment assignment (with standard errors, reported in parentheses, clustered at the unit of randomization). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations. For panel A, columns 5-8 (9-12) report the same entities using the sample of NDC-aligned (non-NDC aligned) households. For panel B, columns 5-8 (9-12) report the same entities on the sample of villages with higher than average (lower than average) baseline support for the NDC. THP sectors are health, water, sanitation, childcare, microcredit; non-THP sectors are road, power, agricultural processing, and primary/secondary education.

Table 5. Poverty Alleviation and Service Access

		ENTIRE SAMPLE				NDC Aligned HHs			Non-NDC Aligned HHs				
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
		ITT	TOT	Control	N	ITT	TOT Effect	Control	N	ITT Effect	TOT	Control	N
		Effect	Effect	mean		Effect	(st. error)	mean		(st. error)	Effect	mean	
		(st. error)	(st. error)	(st. dev.)		(st. error)		(st. dev.)			(st. error)	(st. dev.)	
Overall Well-Being Index		-0.051	-0.097	0.000	2792	-0.232*	-0.430*	-0.050	690	-0.060	-0.128	0.063	1732
		(0.071)	(0.135)	(1.000)		(0.095)	(0.191)	(1.048)		(0.075)	(0.162)	(0.996)	
	Food Security	0.046	0.086	0.000	2749	0.017	0.032	0.123	680	0.045	0.096	-0.042	1707
	Index	(0.046)	(0.087)	(1.000)		(0.076)	(0.139)	(1.170)		(0.051)	(0.109)	(0.952)	
	Literacy and	-0.089	-0.171	0.000	2792	-0.090	-0.167	-0.155	690	-0.120	-0.260	0.057	1732
	Education Index	(0.077)	(0.149)	(1.000)		(0.100)	(0.176)	(1.035)		(0.090)	(0.200)	(1.012)	
	Health and	-0.064	-0.121	0.000	2792	-0.244+	-0.454+	0.026	690	-0.046	-0.099	0.007	1732
	Nutrition Index	(0.087)	(0.166)	(1.000)		(0.144)	(0.273)	(0.950)		(0.083)	(0.178)	(0.993)	
	Water, Env't and	-0.107	-0.199	0.000	2792	-0.250+	-0.460	-0.080	690	-0.096	-0.204	0.085	1732
	Sanitation Index	(0.118)	(0.219)	(1.000)		(0.144)	(0.282)	(1.121)		(0.132)	(0.273)	(0.977)	
Livelihoods and	0.103	0.194	0.000	2792	-0.001	-0.002	-0.037	690	0.078	0.165	0.052	1732	
Financial	(0.087)	(0.160)	(1.000)		(0.115)	(0.207)	(1.061)		(0.095)	(0.194)	(1.008)		
Inclusion Index													

Notes: +significant at 10%; * significant at 5%; ** significant at 1%. Column (1) presents OLS estimates (with standard errors reported in parentheses), clustered at the unit of randomization (village cluster), and controlled for district effects. Each row reports results for a single OLS regression. Column (2) reports IV-GMM treatment-on-the-treated estimates (with standard errors reported in parentheses) with mobilizing to receive an epicenter being the first stage clustered at the unit of randomization (village cluster). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations. Columns 5-8 report the same entities using the sample of NDC-aligned households. Columns 9-12 report the same entities using the sample of non-NDC aligned households. Full details on the construction of each index and the ITT effect and TOT effect on each sub-component are reported in Appendix H.

Did the THP programming, either through the direct results of the programming itself or through its indirect effects on leadership at the community and district level, cause any measurable improvement in the lives of citizens? We measure the aggregate socioeconomic well-being effect of THP by averaging its effects across five broad areas – food security, education and literacy, health and nutrition, environment, and economic livelihoods. We focus on these five outcome areas because they are highlighted in THP’s programming documents and because they are encompassing goals, well-positioned to capture effects even if resources are fungible across sectors, and related closely to the sectors emphasized in the millennium development goals and associated conceptions of human development. For each area of potential impact, we created an index based on variables measuring numerous related outcomes, often combined into sub-indices, as shown in Appendix H.²¹ Collectively, these indices captured specific improvements in well-being in the sectors targeted by THP’s programming – for example, better access to health care, the adoption of specific agricultural practices, and access to credit – as well as broader measures of households’ well-being, such as household income, expenditure and the value of total food consumption.

The effect of THP on the main indices is reported in Table 5. The results indicate that the THP had disappointing results across the entire sample. Focusing on the TOT estimate, THP reduced well-being by 0.10 standard deviations ($se=0.135$), although the effect is imprecisely estimated and

²¹ The construction of the indices was not put forward in a preanalysis plan, as the practice was not common at the time this study began in 2008. However, the survey instrument is available online and provides the basis, without omission, for the construction of the indices. We also based the data collection and thus construction of the indices on indicators emphasized in THP’s own theory of change and programming.

thus particularly large and positive as well as large and negative results cannot be ruled out. However, for NDC-aligned households, the negative effect is starker. Here we estimate a decline in well-being of 0.43 standard deviations as a result of receiving participatory development (se=0.19), which is statistically significant at the 95 percent confidence level. In contrast, for non-NDC aligned households, we cannot reject the null of no effect (effect=-0.13; se=0.16).

How is it that NDC-aligned households in aid-receiving villages became worse off than their counterparts who did not receive participatory development? The evidence in Tables 2 and 3 indicates that NDC-aligned voters were more politically mobilized as a result of participatory development. However, Table 4 suggests that participatory development also caused greater displacement of resources for these households, especially in the allocation of their own household resources but also possibly in the allocation of state resources by local governments. Our interpretation is that NDC supporters were over-mobilized into participatory development: they diverted effort into a project that did not ultimately meet expectations. Importantly, the THP project fell short of expectations in two ways: its direct effects on socioeconomic outcomes through the delivery of public goods and services were smaller than anticipated, and its indirect effects on socioeconomic outcomes through improved engagement with pre-existing political institutions were also negligible, despite the fact that THP was broadly successful in organizing higher levels of engagement.

This provides an explanation for how incumbent co-partisans became distributive losers as a result of participatory development. Revisiting the expectations outlined in Table 1, our explanation is more complex situation than any of the single scenarios included in this table. In

the context of the THP in Ghana, we observe the greater mobilization of government co-partisans theorized in row 3, but this did not translate into increased socioeconomic benefits; in fact, it appears to correspond with effort displacement from more productive activities, leaving co-partisans aid recipients worse off than co-partisans who do not receive aid, as predicted in row 5, column 2.

VI. Conclusion

In a randomized controlled trial of participatory development aid in Ghana, we find high levels of participation from community members, but no change in aggregate socioeconomic outcomes. We also find important heterogeneous treatment effects, specifically that households and villages with pro-government alignment had greater displacement of resources from other efforts towards the new aid-led activities. Yet the project did not end up generating changes in socioeconomic outcomes, thus leading to a negative impact for pro-government households.

As our theoretical expectations outlined in Table 1 indicate, the effects of participatory development aid are complex and likely to vary by context. To think more crisply about the external validity of our findings with respect to participatory development, we highlight two issues related to the specific NGO we study: implementation fidelity and program design. THP's program implementation appears to have delivered on two key objectives regarding process: we observe high levels of participation and inclusiveness. As a result, we think it unlikely that issues with program implementation explain our disappointing findings with regards to investment and socioeconomic outcomes. On program design, THP requires particularly high levels of community

involvement and community contributions compared to other participatory approaches. Although it could be argued this makes it a paradigmatic case of community-based development, it means that displacement effects between involvement in THP and contributions to other local public goods are likely to be particularly pronounced. THP's program is also multi-faceted and multi-sectoral, in contrast to participatory programs that focus on single sectors or provide block grants. By pushing simultaneously in many areas, it may have been more difficult for THP programming to improve upon the outcomes that local contributions and government resources were already accomplishing in these diverse areas. This broad scope of activities could also have created greater implementation challenges.

In this context, we find unintended consequences from participatory development aid, which have implications for both the literature on the cross-institutional effects of participatory development and the literature on the distributional consequences of international aid. We contribute to the first literature by showing the limitations of participatory development even when it meets its goal of encouraging greater participation in government. We find that participatory development institutions can be effective in organizing greater engagement with pre-existing political institutions for co-partisans of the government. In this sense, our study findings contrast with other studies of participatory development; in the few studies that have explicitly examined cross-institutional mobilization by analyzing effects on public participation in local government, scholars have measured null effects (White, Menon, and Waddington 2018), with the one exception being Casey et al. (2012). Our study on the other hand represents a best case scenario for positive cross-institutional effects, and also provides a plausible explanation for the null effects observed elsewhere: in the absence of strong political parties that provide citizens

with access to power, participatory development institutions are likely to have limited effects on political participation.

However, despite this promising first stage effect, the greater mobilization induced by the participatory development institutions did not result in improvements in public goods provision or socioeconomic outcomes. In fact, greater mobilization was actually associated with worse distributional outcomes due to displacement effects. Our findings are striking in that they suggest that even if aid institutions successfully increase mobilization in pre-existing institutions – no small feat – this might not make a difference. In most developing countries, there is limited fiscal decentralization in the sense that that most fiscal power still rests in the national-level executive office, and improved engagement with local representatives may not result in greater local investment (Grossman and Lewis 2014). In fact, in spending more time engaging with relatively powerless local authorities, citizens may be displacing effort from activities that would be more productive in advancing their well-being.

We also contribute to the literature on the distributional consequences of international aid, adding nuance to our understanding of who benefits from aid. To date, the debates in this literature have been on how to ensure aid does not benefit only government supporters. However, we show that – in settings in which aid disappoints – co-partisans of the incumbent government may become distributive losers if they receive aid.

Government copartisans become distributional losers as a result of three aspects of the context studied: the distribution of aid inputs was neutral (both across communities, where it was randomly allocated, and within communities), its effects disappointed relative to initial

expectations, and co-partisans of the government were subject to greater displacement effects. This makes this aid context different from the ones studied by Briggs (2012; 2014) and Jablonski (2014), who observe non-neutral distribution of aid inputs, but do not measure the distributional consequences of aid for socioeconomic outcomes. However, we do not think that the context we study is particularly unique. Indeed, other studies have shown that many aid projects – especially those delivered via NGOs – are delivered neutrally with regards to political cleavages (Brass 2012; Briggs 2017). We also know many aid projects simultaneously fail to have large socioeconomic effects (Easterly 2006; Dionne 2017). We have less existing evidence on how frequently displacement effects are larger for government co-partisans, and this is where we would recommend more research. However, in highly partisan contexts, there are strong theoretical reasons for expecting co-partisans of the government to be subject to greater displacement effects as a result of aid (Golden and Min 2013), suggesting our results may generalize to diverse forms of aid in numerous contexts.

REFERENCES

- Almond, Gabriel, and Sidney Verba. 1963. *The Civic Culture: Political Attitudes and Democracy in Five Nations*. Princeton: Princeton University Press.
- Anderson, Christopher, and Yuliya Tverdova. 2003. "Corruption, Political Allegiances and Attitudes Toward Government in Contemporary Democracies." *American Journal of Political Science* 47 (1): 91–109.
- Arhin, Kwame. 1985. *Traditional Rule in Ghana: Past and Present*. Accra: Sedco Publications.
- Asunka, Joseph. 2017. "Non-Discretionary Resource Allocation as Political Investment: Evidence from Ghana." *Journal of Modern African Studies* 55 (1): 29–53.
- Avdeenko, Alexandra, and Michael Gilligan. 2015. "International Interventions to Build Social Capital: Evidence from A Field Experiment in Sudan." *American Political Science Review* 109 (3): 427–49.
- Baldwin, Kate. 2013. "Why Vote with the Chief? Political Connections and Public Goods Provision in Zambia." *American Journal of Political Science* 57 (4): 794–809.
- Bardhan, Pranab K., and Dilip Mookherjee. 2000. "Capture and Governance at Local and National Levels." *American Economic Review* 90 (2): 135–39.
<https://doi.org/10.1257/aer.90.2.135>.
- Beath, Andrew, Fotini Christia, and Ruben Enikolopov. 2013. "Do Elected Councils Improve Governance? Experimental Evidence on Local Institutions in Afghanistan." *MIT Political Science Department Research Paper* 2013 (24).
- Bergstrom, Theodore, Lawrence Blume, and Hal Varian. 1986. "On the Private Provision of Public Goods." *Journal of Public Economics* 29 (1): 25–49.
- Berman, Sheri. 1997. "Civil Society and Political Institutionalization." *American Behavioral Scientist* 40 (5): 562–74.
- Boulding, Carew. 2010. "NGOs and Political Participation in Weak Democracies: Subnational Evidence on Protest and Voter Turnout from Bolivia." *The Journal of Politics* 72 (April): 456–68. <https://doi.org/10.1017/S0022381609990922>.
- . 2014. *NGOs, Political Protest, and Civil Society*. Cambridge University Press.
- Brass, Jennifer N. 2012. "Why Do NGOs Go Where They Go? Evidence from Kenya." *World Development* 40 (2): 387–401. <https://doi.org/10.1016/j.worlddev.2011.07.017>.
- Briggs, Ryan C. 2012. "Electrifying the Base? Aid and Incumbent Advantage in Ghana." *The Journal of Modern African Studies* 50 (4): 603–24.
<https://doi.org/10.1017/S0022278X12000365>.
- . 2014. "Aiding and Abetting: Project Aid and Ethnic Politics in Kenya." *World Development* 64 (C): 194–205.
- . 2017. "Does Foreign Aid Target the Poorest?" *International Organization* 71 (1): 187–206. <https://doi.org/10.1017/S0020818316000345>.
- Burgess, Robin, Remi Jedwab, Edward Miguel, Ameet Morjaria, and Gerard Padró i Miquel. 2015. "The Value of Democracy: Evidence from Road Building in Kenya." *American Economic Review* 105 (6): 1817–51. <https://doi.org/10.1257/aer.20131031>.
- Bütthe, Tim, Solomon Major, and André de Mello e Souza. 2012. "The Politics of Private Foreign Aid: Humanitarian Principles, Economic Development Objectives, and Organizational

- Interests in NGO Private Aid Allocation." *International Organization* 66 (4): 571–607. <https://doi.org/10.1017/S0020818312000252>.
- Cammett, Melani. 2014. *Compassionate Communalism: Welfare and Sectarianism in Lebanon*. Ithaca: Cornell University Press.
- Casey, Katherine, Rachel Glennerster, and Edward Miguel. 2012. "Reshaping Institutions: Evidence on Aid Impacts Using a Preanalysis Plan." *The Quarterly Journal of Economics* 127 (4): 1755–1812. <https://doi.org/10.1093/qje/qje027>.
- Casey, Katherine, Rachel Glennerster, Edward Miguel, and Maarten Voors. 2018. "Skill versus Voice in Local Development." Working Paper 25022. National Bureau of Economic Research. <https://doi.org/10.3386/w25022>.
- Crook, Richard C., Richard C. Crook, James Manor, and Fellows of the Institute of Development Studies James Manor. 1998. *Democracy and Decentralisation in South Asia and West Africa: Participation, Accountability and Performance*. Cambridge University Press.
- Deserranno, Erika, Aisha Nansamba, and Nancy Qian. 2019. "Aid Crowd-Out: The Effect of NGOs on Government-Provided Public Services," 59.
- Dietrich, Simone. 2013. "Bypass or Engage? Explaining Donor Delivery Tactics in Aid Allocation." *International Studies Quarterly* 57 (4): 698–712.
- Dionne, Kim Yi. 2017. "Doomed Interventions by Kim Yi Dionne." Cambridge Core. November 2017. <https://doi.org/10.1017/9781108157797>.
- Easterly, William. 2006. *The White Man's Burden: Why the West's Efforts to Aid the Rest Have Done so Much Ill and so Little Good*. NY: Penguin.
- Faye, Michael, and Paul Niehaus. 2012. "Political Aid Cycles." *American Economic Review* 102 (7): 3516–30. <https://doi.org/10.1257/aer.102.7.3516>.
- Fearon, James D., Macartan Humphreys, and Jeremy M. Weinstein. 2015. "How Does Development Assistance Affect Collective Action Capacity? Results from a Field Experiment in Post-Conflict Liberia." *American Political Science Review* 109 (3): 450–69. <https://doi.org/10.1017/S0003055415000283>.
- Franck, Raphaël, and Ilia Rainer. 2012. "Does the Leader's Ethnicity Matter? Ethnic Favoritism, Education, and Health in Sub-Saharan Africa." *American Political Science Review* 106 (2): 294–325. <https://doi.org/10.1017/S0003055412000172>.
- Golden, Miriam, and Brian Min. 2013. "Distributive Politics Around the World." *Annual Review of Political Science* 16 (1): 73–99. <https://doi.org/10.1146/annurev-polisci-052209-121553>.
- Grossman, Guy, and Janet I. Lewis. 2014. "Administrative Unit Proliferation." *American Political Science Review* 108 (1): 196–217. <https://doi.org/10.1017/S0003055413000567>.
- Hodler, Roland, and Paul Raschky. 2014. "Regional Favoritism." *The Quarterly Journal of Economics* 129 (2): 995–1033.
- Humphreys, Macartan, Raúl Sánchez de la Sierra, and Peter van der Windt. 2014. "Social and Economic Impacts of Tuungane: Final Report on the Effects of a Community Driven Reconstruction Program in Eastern Democratic Republic of Congo."
- Jablonski, Ryan S. 2014. "How Aid Targets Votes: The Impact of Electoral Incentives on Foreign Aid Distribution." *World Politics* 66 (2): 293–330. <https://doi.org/10.1017/S0043887114000045>.

- Khilji, Nasir, and Ernest Zampelli. 1994. "The Fungibility of U.S. Military and Nonmilitary Assistance and the Impacts on Expenditures of Major Aid Recipients." *Journal of Development Economics* 43: 345–62.
- King, Elisabeth, and Cyrus Samii. 2014. "Fast-Track Institution Building in Conflict-Affected Countries? Insights from Recent Field Experiments." *World Development* 64 (December): 740–54. <https://doi.org/10.1016/j.worlddev.2014.06.030>.
- Kramon, Eric, and Daniel N. Posner. 2013. "Who Benefits from Distributive Politics? How the Outcome One Studies Affects the Answer One Gets." *Perspectives on Politics* 11 (2): 461–74. <https://doi.org/10.1017/S1537592713001035>.
- . 2016. "Ethnic Favoritism in Education in Kenya." *Quarterly Journal of Political Science* 11 (1): 1–58. <https://doi.org/10.1561/100.00015005>.
- Lindberg, Staffan I., and Minion K. C. Morrison. 2005. "Exploring Voter Alignments in Africa: Core and Swing Voters in Ghana." *The Journal of Modern African Studies* 43 (4): 565–86. <https://doi.org/10.1017/S0022278X05001229>.
- Lust, Ellen, and Lise Rakner. 2018. "The Other Side of Taxation: Extraction and Social Institutions in the Developing World." *Annual Review of Political Science* 21 (1): 277–94. <https://doi.org/10.1146/annurev-polisci-042716-102149>.
- MacLean, Lauren M. 2010. *Informal Institutions and Citizenship in Rural Africa*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511730368>.
- Mansuri, Ghazala, and Vijayendra Rao. 2013. *Localizing Development: Does Participation Work?* Washington, D.C: World Bank.
- Mesquita, Bruce Bueno de, and Alastair Smith. 2009. "Political Survival and Endogenous Institutional Change." *Comparative Political Studies* 42 (2): 167–97. <https://doi.org/10.1177/0010414008323330>.
- Moyo, Dambisa. 2009. *Dead Aid: Why Aid Is Not Working and How There Is a Better Way for Africa*. New York: Macmillan.
- Murtazashvili, Jennifer Brick. 2016. *Informal Order and the State in Afghanistan*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781316286890>.
- Nathan, Noah L. 2019. *Electoral Politics and Africa's Urban Transition: Class and Ethnicity in Ghana*. Cambridge Studies in Comparative Politics. Cambridge, United Kingdom ; New York, NY: Cambridge University Press.
- Norris, Pippa. 2011. *Democratic Deficit: Critical Citizens Revisited*. New York: Cambridge University Press.
- Oates, Wallace E. 1972. *Fiscal Federalism*. The Harbrace Series in Business and Economics. New York: Harcourt Brace Jovanovich.
- Olken, Benjamin A., and Monica Singhal. 2011. "Informal Taxation." *American Economic Journal: Applied Economics* 3 (4): 1–28. <https://doi.org/10.1257/app.3.4.1>.
- Ostrom, Elinor. 1990. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge, MA: Cambridge University Press.
- . 1996. "Crossing the Great Divide: Coproduction, Synergy, and Development." *World Development* 24 (6): 1073–87. [https://doi.org/10.1016/0305-750X\(96\)00023-X](https://doi.org/10.1016/0305-750X(96)00023-X).
- Putnam, Robert D, Robert Leonardi, and Raffaella Y Nanetti. 1993. *Making Democracy Work*. Princeton, NJ: Princeton University Press.

- Torpey-Saboe, Nichole. 2015. "Does NGO Presence Decrease Government Spending? A Look at Municipal Spending on Social Services in Brazil." *World Development* 74 (C): 479–88.
- Verba, Sidney, Norman H. Nie, Research Professor of Political Science and Director of the Institute for Quantitative Study of Society Norman H. Nie, Norman H. Nie, and Jae-on Kim. 1978. *Participation and Political Equality: A Seven-Nation Comparison*. Cambridge University Press.
- Weghorst, Keith R., and Staffan I. Lindberg. 2013. "What Drives the Swing Voter in Africa?:" *American Journal of Political Science* 57 (3): 717–34.
<https://doi.org/10.1111/ajps.12022>.
- White, H., R. Menon, and H. Waddington. 2018. "Community-Driven Development: Does It Build Social Cohesion or Infrastructure? A Mixed-Method Evidence Synthesis." Monograph. 2018. <http://www.3ieimpact.org/en/publications/working-papers/3ie-working-paper-30/>.
- Wilson, Louis E. 1987. "The Rise of Paramount Chiefs among the Krobo (Ghana)." *The International Journal of African Historical Studies* 20 (3): 471–95.
<https://doi.org/10.2307/219690>.
- Wright, Joseph, and Matthew Winters. 2010. "The Politics of Effective Foreign Aid." *Annual Review of Political Science* 13 (1): 61–80.
<https://doi.org/10.1146/annurev.polisci.032708.143524>.

On-line Appendices
How Political Insiders Lose Out When International Aid Underperforms:
Evidence from a Participatory Development Experiment in Ghana

Appendix A. Explaining Political Affiliation in Eastern Ghana

Table A1. Correlates of Political Affiliation

	(1)	(2)
	NDC Aligned HH	NDC Aligned HH
Proportion Female	-0.084* (0.042)	-0.087* (0.042)
Average Age	0.000 (0.001)	0.000 (0.001)
Average Education	-0.003 (0.004)	-0.003 (0.003)
Proportion Born in Community	0.028 (0.028)	0.024 (0.028)
Proportion Akwapim	-0.136** (0.029)	-0.077* (0.032)
Proportion Akyem	-0.147** (0.031)	-0.111** (0.035)
Proportion Krobo	0.123** (0.033)	0.075** (0.039)
Proportion Ewe	0.268** (0.036)	0.295** (0.038)
Durable Asset Index	0.006 (0.006)	0.002 (0.005)
Organizational Membership	-0.003 (0.022)	0.005 (0.023)
District Fixed Effects	No	Yes
N	1,796	1,796
R-squared	0.112	0.136

Notes: + significant at 10 %; * significant at 5 %; ** significant at 1 %. Table reports coefficients from OLS regression model with robust standard errors in parentheses below.

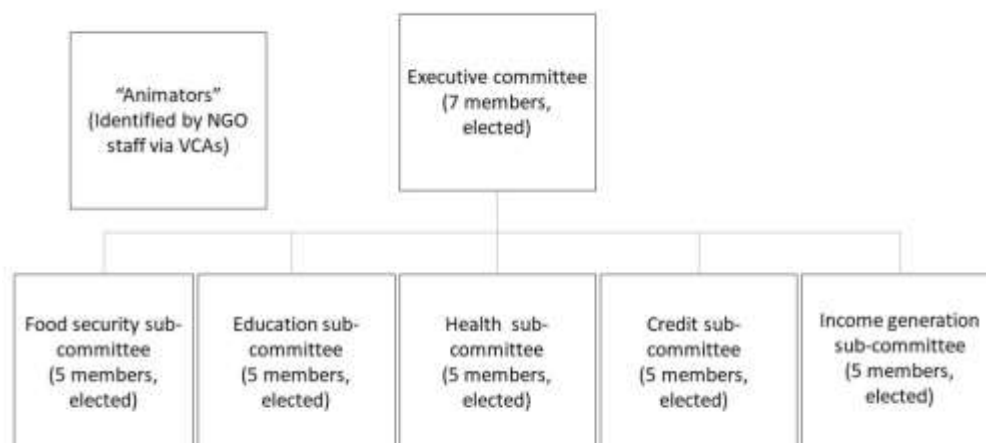
This appendix shows the correlates of households supporting the NDC at the beginning of our study; the outcome variable is whether a majority of adults in the household said they identified with the NDC. This is largely a function of ethnic identity, with households with more Krobo and Ewe members being more likely to identify with the NDC and households with more Akwapim and Akyem members being less likely to do so. In addition, households with more adult women were less likely to identify with the NDC, which likely reflects women's lower levels of partisan mobilization in Ghana.¹

¹ The heterogeneous effects observed by partisanship in the manuscript are not observed when the sample is instead divided by the gender composition of households. (Results available upon request).

Appendix B. The Hunger Project's Participatory Development Approach

This appendix provides further details on The Hunger Project's (THP's) participatory development approach. THP begins its work with communities by organizing "vision, commitment and action" (VCA) workshops in which participants receive training in civic engagement and are encouraged to develop plans to improve their communities. These VCA workshops are repeated regularly throughout the course of the NGO's engagement with a community. Following the initial workshop, two types of leaders are selected to lead programming within their communities: "animators", volunteers identified as having strong leadership skills by the NGO staff who are then asked to help mobilize other community members, and THP committee members, who are elected by the community to oversee programming. Figure B1 illustrates the local leadership structure created as part of the THP process. There is often considerable overlap between animators and committee members, and both sets of leaders subsequently receive further leadership training by the NGO.

Figure B1. THP's participatory development institutions



Once community members demonstrate a commitment to devoting time and resources to collective goods following the initial VCA workshop, THP begins providing financial support for programming activities. At this point, it helps to facilitate the creation of "epicenters," which are community centers containing meeting halls, clinics, rural banks, foodbanks, toilets, a demonstration farm, and either a preschool or library. Once completed, these centers also run agricultural training programs, literacy classes and microfinance programs. THP provides funds to secure the title for the land for the community centers, it hires a contractor to oversee the construction of the center, and it provides some financial support for its education and microfinance programs. However, community members are also expected to devote significant resources in cash or in kind to support the construction of the center, and the goal is to have the local government provide support for many of the programs subsequently run out of the center. Thus, THP's model of change centers mainly around the effects of organizing workshops that develop leadership skills and civic mindedness, not on the effects of a capital infusion

into communities. THP's main emphasis is on engaging new leaders and to forming new community organizations that will help organize future collective activities to benefit the community. In fact, the THP model allows communities only marginal influence over how much resources to devote to different components of the multi-sectoral programming to which THP is committed; this contrasts with community-driven development programs that provide communities with cash grants but is fairly typical of many participatory development programs (Mansuri and Rao 2013; Mosse 2005).²

² For example, in one of our study communities, the committee decided not to build a community center as part of the programming.

Appendix C. Randomization Procedure

THP's model is intended to cater to groups of rural villages with combined populations of about 10,000 people. As a result, in each of the study districts, the research team first determined the communities that were eligible for inclusion in the study – to be eligible, villages had to have populations of less than 2000 people and be situated away from major roads – and then grouped them into village groupings ("clusters") in as naturalistic a way as possible. A public lottery was subsequently held in each district to determine which clusters would be invited to receive THP's programming. The lotteries were conducted by pulling names out of a hat in public, and so no stratification beyond the district level was possible. The lotteries were conducted between September 2006 and September 2008. Due to short-run capacity constraints, THP did not immediately begin engagement with all communities selected for treatment. Within treatment communities, programming was rolled out over a four-year period between 2008-2011.

After the district lotteries, representatives from the communities selected for treatment were invited to participate in a district-level VCA workshop to familiarize themselves with the THP process. The village chief and four other community representatives (2 male, 2 female) from all villages in selected groupings were invited to participate in the workshop.

Appendix D. Balance Statistics and Take-up Analysis

Table D1 shows that we fail to reject that treatment assignment is orthogonal to observable characteristics households and our main outcomes of interest. Each of the variables in this Table is an index. On average, treatment and control households demonstrated similar levels of civic participation and had similar perceptions of their village and district-level leaders. They also showed similar levels of food security, similar health and nutritional access and behaviors, similar access to services related to water, environment and sanitation, and similar economic livelihoods. The only index on which they are statistically significantly different at baseline was literacy and education, with control communities demonstrating higher levels at baseline.

Table D1. Balance Summary Statistics

	(1) Treatment (std dev)	(2) Control (std dev)	(3) Difference (se)	(4) N	(5) Village Took-Up Treatment (std dev)	(6) Village Did Not Take-Up Treatment (std dev)	(7) Difference (se)	(8) N
Community Participation Index	-0.277 (1.208)	-0.278 (1.219)	-0.018 (0.049)	3230	-0.111 (1.236)	-0.436 (1.160)	0.216* (0.072)	1687
Accountability of Village Chief Index	0.408 (1.015)	0.406 (1.018)	0.016 (0.043)	3745	0.393 (1.036)	0.422 (0.992)	0.043 (0.057)	1939
Accountability of District Assemblymember Index	0.452 (1.384)	0.437 (1.431)	-0.001 (0.083)	3647	0.475 (1.370)	0.431 (0.045)	0.030 (0.088)	1897
Food Security Index	-0.955 (0.701)	-0.964 (0.715)	0.002 (0.045)	3645	-0.990 (0.715)	-0.920 (0.684)	-0.143** (0.051)	1903
Literacy and Education Index	-0.201 (0.990)	-0.020 (1.086)	-0.186* (0.078)	3786	-0.321 (0.996)	-0.074 (0.031)	-0.194+ (0.104)	1962
Health and Nutrition Index	0.550 (3.406)	0.487 (1.706)	-0.001 (0.256)	3786	0.658 (4.597)	0.434 (1.212)	0.473 (0.597)	1962
Water, Environment and Sanitation Index	-1.257 (1.751)	-0.952 (1.436)	-0.285 (0.180)	3582	-1.251 (1.864)	-1.263 (1.632)	0.350 (0.241)	1901
Livelihoods and Financial Inclusion Index	-0.080 (1.723)	-0.199 (0.041)	0.118 (0.176)	3786	-0.251 (1.620)	0.015 (1.590)	-0.172 (0.242)	1962
NDC Aligned Household	0.325 (0.442)	0.289 (0.431)	0.027 (0.024)	3267	0.352 (0.449)	0.298 (0.434)	0.023 (0.044)	1707

Notes: + significant at 10 %; * significant at 5 %; ** significant at 1 %. This Table reports baseline summary statistics from the main outcome measures at the household level. Columns (1) and (2) present means (with standard deviations in parentheses) of the treatment and control groups, respectively. Column (3) presents the difference and the standard error of the difference, calculated from an OLS regression model with district fixed effects and standard errors clustered at the unit of randomization (cluster). Column (4) indicates the N. Columns (5) and (6) present means (with standard deviations in parentheses) in the treatment communities that took up the treatment and that did not. Column (7) presents the difference between these communities (calculated as in column (3)), and column (8) indicates the N for this comparison.

Appendix E. Participation in and Governance Structures of THP

Table E1 compares THP leaders to the set of leaders who had ever held traditional leadership positions or held elected office in the village. Specifically, columns (1) through (5) of the table present data on the average (baseline) characteristics of respondents surveyed in our two-wave household survey. Column (1) displays the average characteristics of all adult respondents, column (2) presents the characteristics for respondents who had held a traditional office at some point (mainly village chiefs, subchiefs, linguists, queen mothers and other advisors), column (3) does this for respondents who had held a political office (mainly unit committee members, local party officials, and district assembly members), column (4) shows the characteristics of respondents who had participated in a THP workshop, and column (5) lists the characteristics of respondents who had held leadership positions within THP (animators and committee members). The last three columns of the table show the t-statistic from an unequal t-test comparing (6) the characteristics of all adults to the characteristics of VCA workshop participants; (7) the characteristics of traditional leaders to THP leaders; and (8) the characteristics of political leaders to THP leaders.

The individuals who took part in THP workshops tended to be different from the study communities more broadly. Workshop participants were significantly less likely to be women, significantly older, and significantly more educated than their communities more broadly. On these dimensions, program participants skewed towards those who are already advantaged in existing power structures. Yet, on other dimensions, the program was effective in bringing in disadvantaged community members. In particular, workshop participants were less wealthy (as measured by baseline asset ownership) and more dissatisfied with the president (as measured by trust in the president at baseline) than other community members (though it is noteworthy that they were not more dissatisfied with lower level political and traditional leaders).

In addition, THP managed to create leadership structures that were more inclusive of disadvantaged groups than either traditional institutions or elected institutions within the study communities. THP leaders were more likely to be female than either traditional or political leaders, and they were younger than traditional leaders. Furthermore, like THP workshop participants more generally, they were less wealthy and less aligned with the president at baseline. In this sense, THP's participatory approach appears to have been effective in placing individuals disadvantaged in other governance structures in leadership positions.

Table E1. THP Participants and Leaders Compared to their Communities and Preexisting Leaders

	(1) Mean adults (st. dev)	(2) Mean traditional leaders (st. dev)	(3) Mean political leaders (st. dev)	(4) Mean THP workshop participants (st. dev)	(5) Mean THP leaders (st. dev)	(6) Difference THP workshop vs. adults (st. error)	(7) Difference THP leaders vs. trad/pol. leaders (st. error)
Female	0.529 (0.499) N=2942	0.205 (0.405) N=195	0.110 (0.314) N=100	0.399 (0.491) N=163	0.285 (0.455) N=63	-0.130** (0.040)	0.096 (0.062)
Age (years)	44.5 (17.5) N=2942	55.1 (13.1) N=195	52.6 (12.1) N=100	48.7 (12.6) N=163	50.2 (10.3) N=63	4.2** (1.04)	-4.00** (1.52)
Education (highest grade)	6.18 (4.27) N=2922	7.18 (4.40) N=194	9.31 (2.94) N=98	7.03 (4.11) N=163	8.83 (3.69) N=63	0.85** (0.33)	1.21* (0.53)
Born in village	0.436 (0.496) N=2920	0.407 (0.493) N=194	0.505 (0.503) N=99	0.432 (0.497) N=162	0.503 (0.503) N=63	-0.004 (0.040)	0.070 (0.070)
HH wealth index (baseline)	0.298 (2.100) N=2326	0.502 (2.653) N=157	0.552 (2.254) N=81	0.118 (0.183) N=131	-0.039 (1.820) N=54	-0.180** (0.046)	-0.534+ (0.302)
Organization member (baseline)	0.668 (0.471) N=2779	0.688 (0.465) N=189	0.842 (0.367) N=95	0.826 (0.380) N=161	0.905 (0.296) N=63	0.158** (0.031)	0.114+ (0.068)
NDC supporter (baseline)	0.323 (0.437) N=2533	0.314 (0.440) N=167	0.359 (0.453) N=84	0.367 (0.450) N=142	0.382 (0.454) N=56	0.044 (0.039)	0.061 (0.067)
NPP supporter (baseline)	0.437 (0.458) N=2532	0.483 (0.469) N=167	0.458 (0.457) N=84	0.427 (0.462) N=142	0.347 (0.458) N=56	0.010 (0.039)	-0.121+ (0.069)
Trust chief (baseline)	3.06 (1.06) N=2507	3.16 (0.98) N=168	3.11 (1.04) N=85	3.05 (1.08) N=145	3.06 (1.08) N=58	-0.005 (0.092)	-0.094 (0.157)

Notes: + significant at 10 %; * significant at 5 %; ** significant at 1 %. The first five columns report means, standard deviations (in parentheses) and N for: (1) all adults in treatment villages; (2) all who have held a traditional leadership position in treatment villages; (3) all who have held a political office in treatment villages; (4) all who have participated in a Vision, Commitment and Action workshop run by THP; and (5) all who have served as a leader in the context of THP programming, whether by acting as an animator or a committee member. Column (6) reports the difference in means between the adult population and the participants in the VCA workshops, with the standard error in parentheses. Column (7) reports the difference in means between traditional/political leaders and THP leaders, with the standard errors in parentheses.

Table E2. Exposure to THP Programming

	(1) Treatment Village Take-Up=1 (st. dev.)	(2) Treatment Village Take-Up=0 (st. dev.)	(3) Control mean (st. dev.)	(4) Difference Treatment vs. Control (st. error)	(5) Treatment Village, NDC HH (st. dev.)	(6) Treatment Village, Not NDC HH (st. dev.)	(7) Difference NDC HH vs. Not NDC HH (st. error)
Attended any Vision, Commitment and Action (VCA) session (binary)	0.100 (0.258) N=742	0.013 (0.101) N=665	0.000 (0.000) N=1337	0.058** (0.011)	0.065 (0.220) N=370	0.056 (0.194) N=854	-0.005 (0.035)
Number of VCA sessions attended in last 12 months	0.387 (1.651) N=742	0.030 (0.335) N=665	0.000 (0.000) N=1337	0.213** (0.052)	0.200 (1.143) N=370	0.231 (1.324) N=854	-0.045 (0.081)
Contributed to animator-led project (binary)	0.048 (0.181) N=742	0.011 (0.094) N=665	0.003 (0.044) N=1337	0.026** (0.005)	0.010 (0.070) N=370	0.016 (0.095) N=854	0.001 (0.010)
Attended THP fundraiser (binary)	0.093 (0.251) N=742	0.006 (0.074) N=665	0.001 (0.015) N=1337	0.050** (0.010)	0.017 (0.104) N=370	0.017 (0.101) N=854	-0.016 (0.014)
THP animator (binary)	0.024 (0.112) N=742	0.005 (0.052) N=665	0.000 (0.014) N=1337	0.014** (0.003)	0.028 (0.146) N=370	0.030 (0.146) N=854	-0.007 (0.005)
THP committee member (binary)	0.025 (0.119) N=742	0.007 (0.073) N=665	0.000 (0.000) N=1337	0.016** (0.004)	0.039 (0.172) N=370	0.055 (0.195) N=854	0.000 (0.007)
Any contact with THP programming (binary)	0.381 (0.440) N=742	0.041 (0.178) N=665	0.010 (0.089) N=1337	0.208** (0.034)	0.225 (0.374) N=370	0.195 (0.356) N=854	-0.005 (0.035)
Value of contributions to epicenter and associated programming (cedis)	57.9 (141.4) N=742	7.1 (47.4) N=665	0.8 (13.5) N=1337	30.7** (7.0)	39.5 (120.8) N=370	28.3 (87.3) N=854	8.130 (9.068)

Notes: +significant at 10%; * significant at 5%; ** significant at 1%. The first three columns report means, standard deviations (in parentheses) and N for households in treatment villages that took-up the treatment, households in treatment villages that did not take-up the treatment and households in control villages respectively. Column (4) reports the difference in means between households in villages assigned to treatment and control calculated via OLS regression with district fixed effects and standard errors (reported in parentheses) clustered at the unit of randomization (village cluster). The fifth and sixth columns report means, standard deviations (in parentheses) and N for NDC-aligned households and non-NDC aligned households in villages assigned to treatment. Column (7) reports the difference in means between NDC-aligned and non-NDC aligned households in treatment villages calculated via OLS regression with district fixed effects and standard errors (reported in parentheses) clustered at the unit of randomization (village cluster).

The breadth of inclusion in THP’s programming is also apparent when we examine the proportion of the community included in various aspects of its programming and leadership activities in Table E2. This table begins by comparing the proportion of adults who participated in various THP programs across villages that took up the treatment (column 1) to those that failed to take up the treatment (column 2) and to those in the control group (column 3). The fourth column shows the difference in participation rates across all communities assigned to treatment and all communities assigned to control. Next, column 5 and 6 compare the rate of participation among NDC affiliated households and other households in treatment villages (regardless of take-up), with the seventh column indicating whether there were differences in participation rates based on partisan affiliation.

The first thing to note is that almost no one in the control communities participated in THP’s programming. For each of the programs we consider, the control means approximate zero, and just 1 percent of the adults in the control communities had exposure to any of the programs or activities run by THP. In addition, the very low rates of programming in the communities that failed to take up the treatment suggest that these communities were not significantly exposed to programming after their decline of the invitation to take part. However, large proportions of the adult population participated in THP’s programming in the village groupings that accepted treatment. In these villages, more than 11 percent of adults participated in VCA sessions, almost 10 percent contributed to a THP fundraiser, and 40 percent had participated in some kind of THP programming. THP’s mobilization effort within communities is particularly impressive when one considers participation rates in other community-based development programs; for example, only 0.7 percent of the population is estimated to have participated in village development committee (VDC) member trainings as part of the Tuungane CDD program in the Eastern DRC (Humphreys, Sierra, and Windt 2014).

Appendix F. Attrition Analysis

This appendix examines whether treatment – either by itself or in interaction with baseline outcome variables – affects the likelihood of attrition. We find no evidence of this, as indicated by the F-tests presented at the bottom of the table.

Table F1. Household Attrition

	(1) Completed endline survey	(2) Completed endline survey	(3) Completed endline survey
Treatment	-0.007 (0.018)	-0.004 (0.018)	-0.014 (0.034)
Treatment*Civic participation index			0.022 (0.015)
Treatment*Quality of village leadership index			0.011 (0.014)
Treatment*Perceptions of district leadership index			-0.007 (0.012)
Treatment*Food security index			-0.015 (0.023)
Treatment*Literacy and education index			0.013 (0.018)
Treatment*Health and nutrition index			0.003 (0.005)
Treatment*Environment index			-0.006 (0.012)
Treatment*Livelihoods index			0.010 (0.009)
Treatment*NDC-Aligned HH			-0.019 (0.039)
Control mean	0.742	0.742	0.742
Straight effects for 9 measures	No	Yes	Yes
Treatment interacted with index effects	No	No	Yes
Observations	3786	3786	3786
p-value from F-test that treatment equals zero	0.721	0.817	
p-value from F-test that treatment interacted with indices jointly equals zero			0.684

Notes: + significant at 10%; * significant at 5%; ** significant at 1%. OLS intent-to-treat estimates (with standard errors in parentheses), clustered at the unit of randomization (village cluster). Each column reports results for a single OLS regression of the dependent variables listed in the columns. The dependent variable (non-attrition) is binary, taking 1 if the household was reached for both baseline and endline survey, and 0 if the household was only reached for the baseline and not the endline. All baseline control variables correspond to the outcome variables in Tables 2 & 5, as measured at baseline, with indices standardized to the endline control mean with mean 0 and standard deviation 1. For baseline observations that are missing, the variable is recoded to zero when missing, and a binary indicator of being missing is included into the regression.

Appendix G. Qualitative Data Collection and Results

The statistical analysis of the effects of the NGO's programming is complemented with qualitative evidence collected at two distinct time periods. In 2009, at the beginning of the project roll-out, a research team visited 4 treatment and 4 control villages, conducting multiple in-depth interviews and focus groups at each location. The treatment villages were purposefully selected to include two villages performing well and two villages performing poorly according to The Hunger Project's local staff. The control villages were selected so that they were each from the same district as the treatment villages and of approximately the same size and economic development level.

In 2015, researchers returned to 12 communities (7 treatment, 5 control), again conducting focus groups with citizens and in-depth interviews with community leaders, including individuals who took leadership positions in THP's activities, the elected district assemblyperson and district officials. Seven treatment villages were randomly selected from the districts with earliest exposure to THP in order to trace the effects of THP over the longest duration possible. The selected villages fell in five districts, and we randomly selected one control village in each of these districts for a total of five control villages.

The qualitative interviews found that the socioeconomic results of THP were ultimately disappointing for many participants, who expected larger infusions of capital into their communities. Qualitative interviews conducted in study communities in July 2009 during implementation of the program indicated extremely high expectations for the project, well represented in the following community member's comment: "Looking at how the THP has helped us ... since they arrived, I believe when we work with them, most of our problems will cease."³ However, these initially high expectations had faded by the time the endline interviews were done six years later, with one THP animator noting, "Because they said they were going to alleviate poverty, the community members thought that they were going to give us [more] money." Similarly, a local assemblymember pointed out that "our [community] involvement was very good. With the epicenter for instance we all used our strength to help. When there is something that we have to do, all the community come together to do it...", but the project was not financially sustainable without a greater influx of capital than was received: "We need money to run the activities at the epicenter. This money was not coming from anywhere..."⁴

In addition to the fact that the treated communities received less capital than expected, respondents noted other inefficiencies in THP's service delivery model compared to the local government's model. In particular, they noted the fact that the epicenter buildings were (by design) placed in locations off the main road network or with poor transport connections, making their services more difficult to access than government clinics, even if they were geographically closer as the bird flies.⁵

The promised benefits of greater levels of engagement with pre-existing governing institutions also failed to materialize. Citizens aligned with the NDC did become engaged in politics at all levels as a result

³ Interview with male community member, treatment village, July 2009.

⁴ Interview with THP animator, treatment village, August 2015; interview with assemblymember, treatment village, August 2015.

⁵ Interview with THP animator, treatment village, August 2015.

of THP, which fits with interviewees' emphasis on the importance of partisan connections in mobilizing citizens for a wide range of activities in Ghana's Eastern Region. As one interviewee put it, "If you are a leader and people know your political affiliation and they see that you do not belong to their party, they won't attend communal labor when you call for one. I don't even know what to use to describe partisan politics...If someone knows that you do not belong to his party, he won't even respond to your greetings. It has really affected our relationships negatively." However, even in communities aligned with the incumbent NDC party, the increased levels of engagement with community and district-level government did not translate into more state investment in local public goods and services. In discussing the failure of state support to materialize, interviewees repeatedly noted both that district governments were not very forthcoming in support for the THP projects themselves, aside from sending a nurse to work at the clinic, and the limited influence of elected district assembly members over the local government budget.⁶ In view of the limited political decentralization in Ghana, with the unelected DCE still maintaining a high degree of influence over the district budget, the expectation that better representation could result in better socioeconomic outcomes appears to have been unrealistic.

In view of the ultimately disappointing results of participatory development in this context, some citizens and governments overdisplaced resources from other projects in treated villages. For example, interviewees with budget officers indicated that the government took THP activities into account in developing its own plans in order to avoid duplicating efforts.⁷ But insofar as the THP was not as efficient as the government in providing some services, these communities were harmed by the lack of state investment in these sectors. Importantly, THP projects probably looked particularly successful in NDC-aligned communities, where they generated higher levels of participation in other institutions too. As a result, the local government may have displaced more resources from these projects even without any additional pro-incumbent party bias in local government spending.

⁶ Interview with assemblyman, community 1, August 2015; interview with assemblyman, community 2, August 2015; interview with assemblyman, community 3, August 2015; interview with assemblyman, community 4, August 2015.

⁷ Interview with District Planning Officer, August 2015.

Appendix H. Index Construction and Components

TABLE H1. COMPONENTS OF MAIN POLITICAL INDICES

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) N	(5) Baseline data included in model
Community Participation Index	0.054 (0.045)	0.103 (0.082)	0.000 (1.000)	2746	Yes
Associational membership	0.009 (0.016)	0.016 (0.030)	0.585 (0.430)	2745	No
Attended Last Community Meeting	0.021 (0.019)	0.040 (0.036)	0.472 (0.407)	2746	Yes
Raised Issue at Last Community Meeting	0.018 (0.017)	0.035 (0.032)	0.362 (0.397)	2745	Yes
Village Accountability Index	0.111* (0.047)	0.211* (0.091)	0.000 (1.000)	2744	Yes
Frequency of contact with village chief	0.283* (0.142)	0.539* (0.272)	4.767 (2.292)	2742	No
Extent to which can disagree with village chief	0.046 (0.049)	0.087 (0.093)	2.530 (1.249)	2741	Yes
Trust in village chief	0.087* (0.042)	0.167* (0.082)	3.667 (1.097)	2707	Yes
District Assemblymember Accountability Index	0.069 (0.072)	0.131 (0.131)	0.000 (1.000)	2792	Yes
Frequency of contact with District Assemblymember	0.062 (0.147)	0.118 (0.274)	0.993 (0.086)	2743	No
Satisfaction with District Assemblymember	0.070 (0.052)	0.132 (0.095)	2.089 (0.916)	2742	No
Trust in District Assemblymember	0.059 (0.078)	0.112 (0.144)	2.812 (1.293)	2792	Yes

Notes: +significant at 10%; * significant at 5%; ** significant at 1%. Column (1) presents OLS estimates (with standard errors reported in parentheses), clustered at the unit of randomization (village cluster), and controlled for district effects. Each row reports results for a single OLS regression. Column (2) reports 2SLS treatment-on-the-treated estimates (with standard errors reported in parentheses) with receiving an epicenter being the first stage clustered at the unit of randomization (village cluster). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations and the unit of observation. Column (5) reports whether baseline data is used in the model.

TABLE H2. COMPONENTS OF MAIN SOCIOECONOMIC INDICES

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) N	(5) Baseline data included in model
Food Security Index	0.046 (0.046)	0.046 (0.046)	0.000 (1.000)	2749	Yes
Market price and access improvement (subindex of 2 indicators)	0.032 (0.050)	0.058 (0.092)	0.000 (1.000)	2206	No
Value of food consumption (weekly, GHC)	-4.937* (2.061)	-9.395* (4.118)	73.1 (56.4)	2738	Yes
Agriculture improvements (subindex of 5 indicators)	0.157** (0.057)	0.298** (0.110)	0.000 (1.000)	2739	Yes
Literacy and Education Index	-0.089 (0.077)	-0.171 (0.149)	0.000 (1.000)	2792	Yes
Education (subindex of 2 indicators)	0.005 (0.094)	0.010 (0.178)	0.000 (1.000)	2528	Yes
School quality (subindex of 3 indicators)	-0.116 (0.135)	-0.224 (0.256)	0.000 (1.000)	2116	Yes
Adult literacy/numeracy (subindex of 2 indicators)	-0.060+ (0.033)	-0.113+ (0.064)	0.000 (1.000)	2745	Yes
Female adult literacy/numeracy (subindex of 2 indicators)	-0.069+ (0.039)	-0.130+ (0.075)	0.000 (1.000)	2326	Yes
No child labor	-0.046 (0.063)	-0.086 (0.118)	0.692 (0.462)	2792	Yes
Health and Nutrition Index	-0.064 (0.087)	-0.121 (0.166)	0.000 (1.000)	2792	Yes
Infant survival	-0.002 (0.010)	-0.003 (0.016)	0.993 (0.086)	250	No
Child anthropometry (subindex of 6 indicators)	-0.000 (0.060)	-0.000 (0.109)	0.000 (1.000)	1535	Yes
Health access (subindex of 7 indicators)	-0.088 (0.157)	-0.172 (0.311)	0.000 (1.000)	2792	Yes
Government health services (subindex of 9 indicators)	-0.141 (0.152)	-0.213 (0.223)	0.000 (1.000)	2792	No
Contraception usage	-0.012 (0.027)	-0.022 (0.050)	0.808 (0.385)	1005	No
Prenatal care (subindex of 4 indicators)	-0.034 (0.096)	-0.060 (0.167)	0.000 (1.000)	346	Yes
Postnatal care (subindex of 9 indicators)	-0.362** (0.135)	-0.581** (0.211)	0.000 (1.000)	213	Yes
Number of times immunized	0.308+ (0.163)	0.561+ (0.305)	9.195 (3.039)	1022	Yes
Survival	0.007* (0.003)	0.012* (0.006)	0.972 (0.085)	2792	No
HIV Knowledge (subindex of 4 indicators)	-0.091* (0.041)	-0.173* (0.080)	0.000 (1.000)	2758	Yes

TABLE H2. COMPONENTS OF MAIN SOCIOECONOMIC INDICES (CONTINUED)

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) N	(5) Baseline data included in model
Water, Environment and Sanitation Index	-0.107 (0.118)	-0.199 (0.219)	0.000 (1.000)	2792	Yes
Public sanitation improvements (subindex of 2 indicators)	-0.211+ (0.120)	-0.398+ (0.226)	0.000 (1.000)	2792	Yes
Number of public water facility improvements	-0.074 (0.104)	-0.137 (0.190)	0.859 (0.884)	2686	No
Electricity availability (subindex of 4 indicators)	-0.162 (0.136)	-0.302 (0.257)	0.000 (1.000)	2763	Yes
Agriculture conservation (subindex of 3 indicators)	0.183** (0.058)	0.342** (0.122)	0.000 (1.000)	2418	No
Livelihoods and Financial Inclusion Index	0.103 (0.087)	0.194 (0.160)	0.000 (1.000)	2792	Yes
Enterprise growth (subindex of 4 indicators)	0.022 (0.031)	0.042 (0.057)	0.000 (1.000)	2747	Yes
Durable assets (subindex of 7 indicators)	-0.027 (0.050)	-0.052 (0.094)	0.000 (1.000)	2750	Yes
Farm investment (annual, GHC)	26.412 (71.389)	49.304 (132.695)	557.4 (1287.1)	2396	Yes
Household income (annual, GHC)	-59415.6 (39428.5)	-113612.9 (75177.1)	70222.8 (1710983.8)	2750	Yes
Financial inclusion - savings (subindex of 5 indicators)	0.062 (0.125)	0.116 (0.228)	0.000 (1.000)	2792	Yes
Financial inclusion - credit (subindex of 4 indicators)	0.294* (0.131)	0.556* (0.237)	0.000 (1.000)	2792	Yes
Non-food household expenditures (monthly, GHC)	6.740 (16.902)	12.793 (31.685)	531.1 (438.3)	2741	Yes

Notes: +significant at 10%; * significant at 5%; ** significant at 1%. Column (1) presents OLS estimates (with standard errors reported in parentheses), clustered at the unit of randomization (village cluster), and controlled for district effects. Each row reports results for a single OLS regression. Column (2) reports 2SLS treatment-on-the-treated estimates (with standard errors reported in parentheses) with receiving an epicenter being the first stage clustered at the unit of randomization (village cluster). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations and the unit of observation. Column (5) reports whether baseline data is used in the model.

TABLE H3. COMPONENTS OF MAIN SOCIOECONOMIC INDICES, NDC ALIGNED HHs

	ITT Effect (standard error)	TOT Effect (standard error)	Control mean (standard dev.)	N	Baseline data included in model
Food Security Index	0.017 (0.076)	0.032 (0.140)	0.131 (1.69)	680	Yes
Market price and access improvement (subindex of 2 indicators)	0.078 (0.126)	0.140 (0.157)	0.126 (1.322)	550	No
Value of food consumption (weekly, GHC)	-9.979* (4.196)	-18.545* (7.994)	77.6 (69.5)	679	Yes
Agriculture improvements (subindex of 5 indicators)	0.146 (0.106)	0.272 (0.205)	0.060 (1.086)	680	Yes
Literacy and Education Index	-0.090 (0.099)	-0.167 (0.176)	-0.155 (1.035)	690	Yes
Education (subindex of 2 indicators)	0.123 (0.134)	0.235 (0.257)	-0.156 (0.991)	618	Yes
School quality (subindex of 3 indicators)	-0.285** (0.106)	-0.632** (0.215)	0.111 (0.687)	441	Yes
Adult literacy/numeracy (subindex of 2 indicators)	-0.090 (0.072)	-0.167 (0.132)	-0.128 (0.969)	681	Yes
Female adult literacy/numeracy (subindex of 2 indicators)	-0.134+ (0.068)	-0.244* (0.123)	-0.116 (0.974)	576	Yes
No child labor	0.010 (0.064)	0.018 (0.116)	0.685 (0.465)	690	Yes
Health and Nutrition Index	-0.244+ (0.144)	-0.454+ (0.273)	0.026 (0.950)	690	Yes
Infant survival	-0.032 (0.035)	-0.057 (0.059)	1.000 (0.000)	76	No
Child anthropometry (subindex of 6 indicators)	0.009 (0.102)	0.017 (0.179)	0.006 (0.976)	396	Yes
Health access (subindex of 7 indicators)	-0.063 (0.182)	-0.122 (0.348)	-0.083 (0.977)	690	Yes
Government health services (subindex of 9 indicators)	-0.298 (0.259)	-0.435 (0.358)	0.197 (1.229)	380	No
Contraception usage	-0.002 (0.037)	0.012 (0.067)	0.798 (0.388)	238	No
Prenatal care (subindex of 4 indicators)	-0.437+ (0.250)	-0.655+ (0.381)	0.069 (0.894)	95	Yes
Postnatal care (subindex of 9 indicators)	-0.213 (0.284)	-0.322 (0.318)	0.068 (1.011)	66	Yes
Number of times immunized	0.586+ (0.347)	0.981+ (0.582)	8.915 (3.237)	278	Yes
Survival	-0.005 (0.006)	-0.009 (0.012)	0.975 (0.071)	690	Yes
HIV Knowledge (subindex of 4 indicators)	-0.196* (0.090)	-0.363* (0.168)	-0.065 (0.993)	681	Yes

TABLE H3. COMPONENTS OF MAIN SOCIOECONOMIC INDICES, NDC ALIGNED HHs (CONTINUED)

	ITT Effect (standard error)	TOT Effect (standard error)	Control mean (standard dev.)	N	Baseline data included in model
Water, Environment and Sanitation Index	-0.250+ (0.144)	-0.460 (0.282)	-0.080 (1.121)	690	Yes
Public sanitation improvements (subindex of 2 indicators)	-0.350** (0.125)	-0.650* (0.272)	-0.091 (1.122)	690	Yes
Number of public water facility improvements	-0.181 (0.144)	-0.332 (0.267)	0.855 (0.951)	661	No
Electricity availability (subindex of 4 indicators)	-0.281+ (0.167)	-0.511 (0.316)	-0.067 (1.026)	679	Yes
Agriculture conservation (subindex of 3 indicators)	0.136 (0.086)	0.248 (0.161)	0.056 (0.973)	609	No
Livelihoods and Financial Inclusion Index	-0.001 (0.115)	-0.002 (0.207)	-0.037 (1.061)	690	Yes
Enterprise growth (subindex of 4 indicators)	0.096 (0.066)	0.179 (0.122)	-0.107 (1.018)	680	Yes
Durable assets (subindex of 7 indicators)	0.016 (0.044)	0.029 (0.081)	-0.157 (0.547)	681	Yes
Farm investment (annual, GHC)	67.7 (80.0)	122.75 (146.42)	487.2 (900.6)	608	Yes
Household income (annual, GHC)	-201702.8 (183863.4)	-376114.3 (341373.3)	188579.5 (3105460.7)	681	Yes
Financial inclusion - savings (subindex of 5 indicators)	0.000 (0.137)	0.000 (0.248)	-0.004 (0.952)	690	Yes
Financial inclusion - credit (subindex of 4 indicators)	0.069 (0.157)	0.126 (0.278)	0.048 (0.972)	690	Yes
Non-food household expenditures (monthly, GHC)	-53.1+ (29.0)	-98.8+ (53.7)	561.4 (406.7)	690	Yes

Notes: *significant at 10%; ** significant at 5%; *** significant at 1%. Column (1) presents OLS estimates (with standard errors reported in parentheses), clustered at the unit of randomization (village cluster), and controlled for district effects. Each row reports results for a single OLS regression. Column (2) reports 2SLS treatment-on-the-treated estimates (with standard errors reported in parentheses) with receiving an epicenter being the first stage clustered at the unit of randomization (village cluster). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations and the unit of observation. Column (5) reports whether baseline data is used in the model.

TABLE H4. COMPONENTS OF MAIN SOCIOECONOMIC INDICES, NON-NDC ALIGNED HHs

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) N	(5) Baseline data included in model
Food Security Index	0.045 (0.051)	0.096 (0.109)	-0.042 (0.952)	1707	Yes
Market price and access improvement (subindex of 2 indicators)	0.032 (0.056)	0.066 (0.118)	-0.072 (0.870)	1361	No
Value of food consumption (weekly, GHC)	-2.890 (2.194)	-6.165 (4.893)	71.6 (52.8)	1699	Yes
Agriculture improvements (subindex of 5 indicators)	0.126* (0.063)	0.268* (0.132)	0.003 (0.995)	1700	Yes
Literacy and Education Index	-0.120 (0.090)	-0.260 (0.199)	0.057 (1.012)	1732	Yes
Education (subindex of 2 indicators)	-0.040 (0.114)	-0.084 (0.237)	0.000 (1.013)	1579	Yes
School quality (subindex of 3 indicators)	-0.217+ (0.127)	-0.464+ (0.276)	0.098 (0.973)	1368	Yes
Adult literacy/numeracy (subindex of 2 indicators)	-0.070+ (0.036)	-0.150+ (0.080)	0.069 (1.010)	1703	Yes
Female adult literacy/numeracy (subindex of 2 indicators)	-0.078+ (0.043)	-0.164+ (0.090)	0.062 (1.014)	1437	Yes
No child labor	-0.029 (0.072)	-0.062 (0.153)	0.697 (0.460)	1732	Yes
Health and Nutrition Index	-0.046 (0.083)	-0.099 (0.178)	0.007 (0.994)	1732	Yes
Infant survival	0.011 (0.012)	0.020 (0.020)	0.985 (0.121)	142	No
Child anthropometry (subindex of 6 indicators)	0.024 (0.076)	0.050 (0.157)	-0.020 (0.997)	944	Yes
Health access (subindex of 7 indicators)	-0.128 (0.180)	-0.286 (0.414)	0.031 (1.028)	1732	Yes
Government health services (subindex of 9 indicators)	-0.116 (0.150)	-0.192 (0.240)	-0.053 (0.916)	1166	Yes
Contraception usage	-0.039 (0.033)	-0.077 (0.067)	0.818 (0.377)	645	No
Prenatal care (subindex of 4 indicators)	0.109 (0.118)	0.219 (0.220)	-0.045 (1.036)	200	Yes
Postnatal care (subindex of 9 indicators)	-0.406* (0.177)	-0.713* (0.293)	-0.077 (0.952)	120	Yes
Number of times immunized	0.165 (0.184)	0.339 (0.376)	9.421 (2.856)	609	Yes
Survival	0.006 (0.004)	0.013 (0.008)	0.974 (0.083)	1732	No
HIV Knowledge (subindex of 4 indicators)	-0.061 (0.053)	-0.131 (0.116)	0.011 (1.028)	1714	Yes

TABLE H4. COMPONENTS OF MAIN SOCIOECONOMIC INDICES, NON-NDC ALIGNED HHs (CONTINUED)

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) N	(5) Baseline data included in model
Water, Environment and Sanitation Index	-0.096 (0.132)	-0.204 (0.273)	0.085 (0.977)	1732	Yes
Public sanitation improvements (subindex of 2 indicators)	-0.175 (0.135)	-0.374 (0.283)	0.046 (0.996)	1732	Yes
Number of public water facility improvements	-0.035 (0.119)	-0.073 (0.245)	0.922 (0.903)	1660	No
Electricity availability (subindex of 4 indicators)	-0.200 (0.147)	-0.421 (0.311)	0.073 (1.006)	1716	Yes
Agriculture conservation (subindex of 3 indicators)	0.173* (0.071)	0.365* (0.161)	0.008 (1.041)	1487	No
Livelihoods and Financial Inclusion Index	0.078 (0.095)	0.165 (0.194)	0.052 (1.008)	1732	Yes
Enterprise growth (subindex of 4 indicators)	-0.024 (0.046)	-0.052 (0.099)	0.057 (0.955)	1705	Yes
Durable assets (subindex of 7 indicators)	-0.058 (0.065)	-0.125 (0.136)	0.070 (1.156)	1707	Yes
Farm investment (annual, GHC)	-1.915 (85.8)	-4.026 (178.5)	593.1 (1480.7)	1474	Yes
Household income (annual, GHC)	-33372.7 (31162.5)	-71322.2 (65934.2)	41033.6 (1045519.2)	1707	Yes
Financial inclusion - savings (subindex of 5 indicators)	0.014 (0.151)	0.028 (0.312)	0.044 (1.077)	1732	Yes
Financial inclusion - credit (subindex of 4 indicators)	0.332* (0.140)	0.702* (0.273)	0.004 (1.048)	1732	Yes
Non-food household expenditures (monthly, GHC)	21.5 (20.9)	45.6 (43.6)	523.9 (448.4)	1701	Yes

Notes: +significant at 10%; * significant at 5%; ** significant at 1%. Column (1) presents OLS estimates (with standard errors reported in parentheses), clustered at the unit of randomization (village cluster), and controlled for district effects. Each row reports results for a single OLS regression. Column (2) reports 2SLS treatment-on-the-treated estimates (with standard errors reported in parentheses) with receiving an epicenter being the first stage clustered at the unit of randomization (village cluster). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations. Column (5) reports whether baseline data was included in the model.

TABLE H5. COMPONENTS OF SUBINDICES

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) No. HHs	(5) No. Villages	(5) Baseline data included in model	(6) Level of data collection
Market price and access improvement subindex	0.032 (0.050)	0.058 (0.092)	0.000 (1.000)	2206	194	No	
Maize market price (GHC)	-55.4 (41.2)	-105.179 (78.316)	136.3 (1103.8)	1048	187	No	Household
Sold maize (binary)	0.030 (0.029)	0.056 (0.054)	0.476 (0.540)	2206	194	No	Household
Agriculture improvements subindex	0.157** (0.057)	0.298** (0.110)	0.000 (1.000)	2739	194	Yes	
Number of farm improvements	0.286** (0.082)	0.535** (0.168)	1.165 (1.421)	2418	194	No	Household
Farm output market value (annual, GHC)	121.9 (241.4)	221.861 (433.276)	2294.3 (5491.3)	2126	192	Yes	Household
Number of cultivated acres	0.242 (0.396)	0.452 (0.733)	5.029 (12.2)	2412	194	No	Household
Current livestock value (GHC)	272.1 (179.3)	510.514 (346.837)	791.8 (1941.5)	2251	194	No	Household
Number of types of livestock owned	0.088 (0.054)	0.167 (0.103)	1.480 (1.085)	2738	194	No	Household
Education subindex	0.005 (0.094)	0.010 (0.178)	0.000 (1.000)	2528	194	Yes	
Highest number of years of education	-0.039 (0.129)	-0.071 (0.234)	3.322 (2.794)	2004	194	Yes	Household
Average school attendance percentage in community	0.015 (0.013)	0.029 (0.026)	0.822 (0.089)	1938	132	Yes	Village
School quality subindex	-0.116 (0.135)	-0.224 (0.256)	0.000 (1.000)	2116	144	Yes	
Hours in school day	-0.149+ (0.088)	-0.345 (0.197)	6.460 (0.862)	1695	115	Yes	Village
Years of education of instructors	0.326 (0.244)	0.704 (0.548)	14.552 (1.562)	1882	129	Yes	Village
Teacher-student ratio	-0.048 (0.032)	-0.073 (0.061)	0.101 (0.245)	1890	129	Yes	Village
Adult literacy/numeracy subindex	-0.060+ (0.033)	-0.113+ (0.064)	0.000 (1.000)	2745	194	Yes	
Literate (binary)	-0.021+ (0.012)	-0.040 (0.024)	0.439 (0.385)	2745	194	Yes	Individual
Numerator (binary)	-0.019 (0.014)	-0.036 (0.027)	0.623 (0.381)	2745	194	Yes	Individual
Female adult literacy/numeracy subindex	-0.069+ (0.039)	-0.130+ (0.075)	0.000 (1.000)	2326	194	Yes	
Literate (binary)	-0.036* (0.016)	-0.068* (0.031)	0.319 (0.408)	2326	194	Yes	Individual
Numerator (binary)	-0.016 (0.018)	-0.031 (0.034)	0.520 (0.448)	2326	194	Yes	Individual

TABLE H5. COMPONENTS OF SUBINDICES (CONTINUED, PAGE 2)

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Congtrol mean (standard dev.)	(4) No. HHs	(5) No. Villages	(5) Baseline data included in model	(6) Level of data collection
Child anthropometry subindex	-0.000 (0.060)	-0.000 (0.109)	0.000 (1.000)	1535	194	Yes	
Height (cm), age 2 through 5	-0.995 (0.990)	-1.801 (1.796)	96.9 (12.4)	821	186	No	Individual
Weight (kg), age 2 through 5	-0.090 (0.239)	-0.163 (0.426)	12.6 (3.247)	821	186	Yes	Individual
Arm circumference (cm), age 2 through 5	-0.064 (0.109)	0.118 (0.199)	15.8 (1.833)	819	186	Yes	Individual
Height (cm), age 6 through 12	0.972 (1.083)	1.807 (2.011)	124.6 (17.8)	1315	193	Yes	Individual
Weight (kg), age 6 through 12	0.284 (0.381)	0.524 (0.704)	23.2 (6.926)	1315	193	Yes	Individual
Arm circumference (cm), age 6 through 12	0.049 (0.139)	0.091 (0.257)	18.2 (2.289)	1315	193	Yes	Individual
Health access subindex	-0.088 (0.157)	-0.172 (0.311)	0.000 (1.000)	2792	194	Yes	
Health center built since 2008	0.043 (0.066)	0.081 (0.123)	0.159 (0.366)	2792	194	No	Village
Number of types of immunizations available in nearest health center	-0.788+ (0.434)	-1.297+ (0.730)	6.072 (1.633)	1721	116	Yes	Village
Number of average patients (daily) treated in nearest health center	-5.538 (4.355)	-8.042 (6.451)	23.9 (23.3)	1690	114	Yes	Village
Prenatal care availability in nearest health center (binary)	-0.040 (0.069)	-0.076 (0.110)	0.853 (0.354)	1745	118	Yes	Village
Delivery availability in nearest health center (binary)	0.013 (0.097)	0.017 (0.151)	0.573 (0.495)	1745	118	Yes	Village
Number of beds in nearest health center	0.188 (0.658)	0.317 (0.969)	3.047 (3.554)	1676	113	Yes	Village
Number of days per week head of nearest health center works	0.358+ (0.204)	0.558+ (0.324)	6.200 (1.115)	1734	117	No	Village

TABLE H5. COMPONENTS OF SUBINDICES (CONTINUED, PAGE 3)

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) No. HHs	(5) No. Villages	(5) Baseline data included in model	(6) Level of data collection
Government health services subindex	-0.141 (0.152)	-0.213 (0.223)	0.000 (1.000)	1717	116	No	
Frequency of visits to chlorinate wells (0 = never, 7 = once a week)	-0.143 (0.226)	-0.214 (0.329)	0.566 (1.460)	1702	115	No	Village
Frequency of visits to provide malaria eradication services (0 = never, 7 = once a week)	0.181 (0.394)	0.286 (0.595)	2.006 (2.207)	1717	116	No	Village
Frequency of visits to provide pre- and post-natal care (0 = never, 7 = once a week)	-0.199 (0.411)	-0.301 (0.611)	1.402 (2.138)	1717	116	No	Village
Frequency of visits to provide nutritional supplements (0 = never, 7 = once a week)	0.010 (0.360)	0.014 (0.539)	0.813 (1.785)	1717	116	No	Village
Frequency of visits to provide general health education (0 = never, 7 = once a week)	-0.313 (0.360)	-0.471 (0.535)	1.926 (2.259)	1717	116	No	Village
Frequency of visits to provide family planning education (0 = never, 7 = once a week)	-0.408 (0.411)	-0.617 (0.592)	2.044 (2.331)	1717	116	No	Village
Frequency of visits to distribute condoms (0 = never, 7 = once a week)	-0.373 (0.331)	-0.565 (0.484)	1.020 (1.973)	1717	116	No	Village
Frequency of visits to provide HIV/AIDS education (0 = never, 7 = once a week)	-0.836* (0.396)	-1.266* (0.626)	1.859 (2.272)	1717	116	No	Village
Frequency of visits to provide guinea worm education & eradication (0 = never, 7 = once a week)	-0.087 (0.372)	-0.133 (0.559)	2.049 (2.433)	1706	115	No	Village
Prenatal care subindex	-0.034 (0.096)	-0.060 (0.167)	0.000 (1.000)	346	162	Yes	
Received some prenatal care (binary)	-0.002 (0.035)	-0.003 (0.061)	0.839 (0.366)	346	162	Yes	Individual
Earliness of prenatal care ((40-week of pregnancy in which prenatal care began)/40)	-0.014 (0.027)	-0.024 (0.048)	0.627 (0.308)	344	162	Yes	Individual
Went to a "good" prenatal practitioner (binary)	-0.003 (0.036)	-0.006 (0.062)	0.839 (0.366)	346	162	Yes	Individual
Number of times went to prenatal care	-0.259 (0.348)	-0.456 (0.614)	4.716 (3.434)	346	162	Yes	Individual

TABLE H5. COMPONENTS OF SUBINDICES (CONTINUED, PAGE 4)

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) No. HHs	(5) No. Villages	(5) Baseline data included in model	(6) Level of data collection
Postnatal care subindex	-0.362** (0.135)	-0.581** (0.211)	0.000 (1.000)	213	213	Yes	
Received some postnatal care (binary)	-0.040 (0.039)	-0.065 (0.060)	0.900 (0.298)	131	213	Yes	Individual
Number of times went to postnatal care	-0.382 (0.595)	-0.605 (0.903)	4.752 (4.250)	131	213	Yes	Individual
Child breastfed (binary)	-0.009 (0.006)	-0.014 (0.010)	1.000 (0.000)	131	213	No	Individual
Child not given water before 6 months (binary)	-0.065 (0.067)	-0.104 (0.101)	0.643 (0.481)	130	212	No	Individual
Child not given liquid before 6 months (binary)	-0.106* (0.052)	-0.170* (0.079)	0.757 (0.431)	130	212	No	Individual
Child not given solid food before 6 months (binary)	-0.031 (0.032)	-0.052 (0.048)	0.956 (0.206)	129	211	No	Individual
Height (cm), age < 2	-3.011+ (1.765)	-4.522+ (2.575)	64.3 (15.3)	128	196	No	Individual
Weight (kg), age < 2	-0.565+ (0.335)	-0.857+ (0.487)	7.461 (2.485)	128	197	Yes	Individual
Arm circumference (cm), age < 2	-0.040 (0.261)	-0.139 (0.373)	14.0 (1.701)	128	197	Yes	Individual
HIV Knowledge subindex	-0.091* (0.041)	-0.173* (0.080)	0.000 (1.000)	2758	194	Yes	
Heard of HIV (binary)	-0.017* (0.007)	-0.033* (0.014)	0.931 (0.171)	2758	194	Yes	Individual
Number of accurate ways known to prevent HIV (max 3)	-0.059* (0.026)	-0.113* (0.051)	1.466 (0.658)	2758	194	Yes	Individual
Knew that a person with HIV could still look healthy (binary)	-0.009 (0.014)	-0.017 (0.026)	0.743 (0.337)	2758	194	Yes	Individual
Knew that HIV can be transmitted from mother to child (binary)	-0.015 (0.012)	-0.029 (0.023)	0.719 (0.332)	2758	194	Yes	Individual
Public sanitation improvements subindex	-0.211+ (0.120)	-0.398+ (0.226)	0.000 (1.000)	2792	194	Yes	
Number of improvements made to any public sanitation facilities in community	-0.206 (0.135)	-0.359 (0.239)	0.689 (1.033)	2493	174	No	Village
Number of good sanitation practices visible in community	-0.178* (0.080)	-0.325* (0.152)	5.806 (0.540)	2754	192	No	Village

TABLE H5. COMPONENTS OF SUBINDICES (CONTINUED, PAGE 5)

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) No. HHs	(5) No. Villages	(5) Baseline data included in model	(6) Level of data collection
Electricity availability subindex	-0.162 (0.136)	-0.302 (0.257)	0.000 (1.000)	2763	192	Yes	
Electricity from main grid available in community (binary)	-0.049 (0.054)	-0.092 (0.101)	0.463 (0.499)	2763	192	Yes	Village
Electricity established in past 5 years (binary)	-0.021 (0.089)	-0.035 (0.148)	0.355 (0.479)	1152	74	Yes	Village
Percentage of households connected to electricity	0.034 (4.112)	0.063 (7.510)	31.9 (37.6)	2763	192	Yes	Village
Number of days per month with no loss of electricity from more than 3 hrs	-0.378 (0.896)	-1.064 (1.513)	24.4 (5.371)	1153	74	Yes	Village
Agriculture conservation subindex	5.405 (4.671)	10.1 (8.939)	9.273 (56.9)	2416	194	No	Household
Number of agricultural improvements to farm made in past year	0.033* (0.013)	0.061* (0.025)	0.067 (0.282)	2417	194	No	Household
Number of trees planted	0.022 (0.031)	0.042 (0.057)	0.000 (1.000)	2747	194	Yes	
Soil-enriching legumes planted	-44.2 (42.7)	-80.3 (79.0)	207.7 (932.6)	1297	192	Yes	Household
Enterprise growth subindex	0.048 (0.138)	0.088 (0.247)	4.533 (2.100)	1324	192	No	Household
Business profit (monthly, GHC)	-0.039 (0.103)	-0.070 (0.186)	1.501 (2.854)	1326	192	No	Household
Number of days per week business runs	0.011 (0.009)	0.021 (0.016)	0.893 (0.275)	2745	194	Yes	Individual
Number of workers at business	-0.027 (0.050)	-0.052 (0.094)	0.000 (1.000)	2750	194	Yes	
Belief that a new business can be worth the investment (binary)	-0.010 (0.036)	-0.018 (0.068)	0.113 (0.486)	2750	194	Yes	Household

TABLE H5. COMPONENTS OF SUBINDICES (CONTINUED, PAGE 6)

	(1) ITT Effect (standard error)	(2) TOT Effect (standard error)	(3) Control mean (standard dev.)	(4) No. HHs	(5) No. Villages	(5) Baseline data included in model	(6) Level of data collection
Durable assets subindex	-0.015 (0.017)	-0.029 (0.032)	0.073 (0.434)	2750	194	Yes	Household
Number of TVs owned	-0.001 (0.022)	-0.003 (0.041)	0.131 (0.434)	2750	194	Yes	Household
Number of satellites owned	-0.034 (0.030)	-0.065 (0.057)	0.192 (0.570)	2750	194	Yes	Household
Number of refrigerators owned	-0.006 (0.017)	-0.012 (0.033)	0.171 (0.478)	2750	194	Yes	Household
Number of electric fans owned	0.013 (0.014)	0.025 (0.026)	0.013 (0.193)	2750	194	Yes	Household
Number of sewing machines owned	-0.013 (0.025)	-0.026 (0.047)	0.223 (0.588)	2750	194	Yes	Household
Number of motorcycles owned	0.062 (0.125)	0.116 (0.228)	0.000 (1.000)	2792	194	Yes	
Number of bicycles owned	0.006 (0.021)	0.012 (0.039)	0.361 (0.480)	2792	194	Yes	Household
Financial inclusion – savings subindex	189.0 (237.1)	349.6 (435.1)	956.0 (2757.4)	1024	189	Yes	Household
Has savings (binary)	-37.3 (136.4)	-67.7 (245.0)	589.6 (1954.0)	984	189	Yes	Household
Savings flow (yearly, GHC)	0.018 (0.037)	0.033 (0.068)	0.045 (0.208)	2792	194	Yes	Village
Savings balance (GHC)	-37.3 (136.4)	-67.7 (245.0)	589.6 (1954.0)	984	189	Yes	Household
Existence of local financial institution	0.018 (0.037)	0.033 (0.068)	0.045 (0.208)	2792	194	Yes	Village
Financial inclusion – credit subindex	0.294* (0.131)	0.556* (0.237)	0.000 (1.000)	2792	194	Yes	
Formal borrowing, past year (binary)	0.028+ (0.015)	0.053+ (0.027)	0.072 (0.259)	2746	194	Yes	Household
Amount of formal loan, past year	14.9 (18.7)	28.3 (35.7)	57.4 (362.2)	2746	194	Yes	Household
Local institution provides loans	0.041 (0.032)	0.077 (0.058)	0.014 (0.118)	2792	194	Yes	Village
100 - interest rate at local financial institution	2.917* (1.362)	6.567* (2.445)	69.9 (11.5)	760	52	No	Village

Notes: +significant at 10%; * significant at 5%; ** significant at 1%. Column (1) presents OLS estimates (with standard errors reported in parentheses), clustered at the unit of randomization (village cluster), and controlled for district effects. Each row reports results for a single OLS regression. Column (2) reports 2SLS treatment-on-the-treated estimates (with standard errors reported in parentheses) with receiving an epicenter being the first stage clustered at the unit of randomization (village cluster). Column (3) reports endline control means (with standard deviations reported in parentheses). Column (4) reports the number of observations. Column (5) reports the number of villages. Column (6) reports whether baseline data is used in the model. Column (7) reports the level of measurement.

Appendix I. First-Stage of Instrumental Variable Results

This appendix shows a strong first stage effect of assignment to treatment on the probability of a village mobilizing to receive participatory programming.

Table I1. TOT first stage regression

	(1) Mobilized
Treatment	0.530** (0.069)
N	2792

Notes: + significant at 10%; * significant at 5%; ** significant at 1%. Treatment is defined as having received an invitation to mobilize the community to build an epicenter. Standard errors, clustered at the unit of randomization (village cluster), are reported in parentheses. The first stage is calculated using OLS with district fixed effects. The unit of observation is the household.

Appendix J. Village-level results by different partisan cut-offs

This appendix shows that electoral area-level results presented in Table 3 and 4 of the manuscript are not dependent on the specific cut-off used to defined NDC-aligned electoral areas (30 %). At the 30% cut-off, there are 50 NDC-aligned electoral areas (44 %) and 64 non-aligned electoral areas (55%) in our sample. If we define NDC-aligned electoral areas as those where at least 25 % of HHs are NDC-aligned at baseline, then we have 63 NDC-aligned electoral areas (55%) and 51 non-NDC aligned electoral areas (45%). If we define NDC-aligned electoral areas as those where at least 35 % of HHs are NDC-aligned at baseline, then we have 40 NDC-aligned electoral areas (35%) and 74 non-NDC aligned electoral areas (65%).

Figures J1, J2, J3, J4, J5 and J6 plot the ITT estimates for non-NDC electoral areas and NDC-electoral areas respectively for each of the electoral-area outcomes considered in Table 3 and 4 by the three different definitions of NDC-aligned electoral areas. Overall, the results are very consistent regardless of the cut-off used to define NDC-alignment. In only one instance does the interpretation of the results depend on the cut-off used to define NDC-alignment; we no longer observe greater activity levels by local representatives in NDC-aligned villages when using the demanding 35% threshold for defining NDC-aligned villages.

Figure J1. Turnout across non-NDC and NDC Aligned Villages by different cutoffs

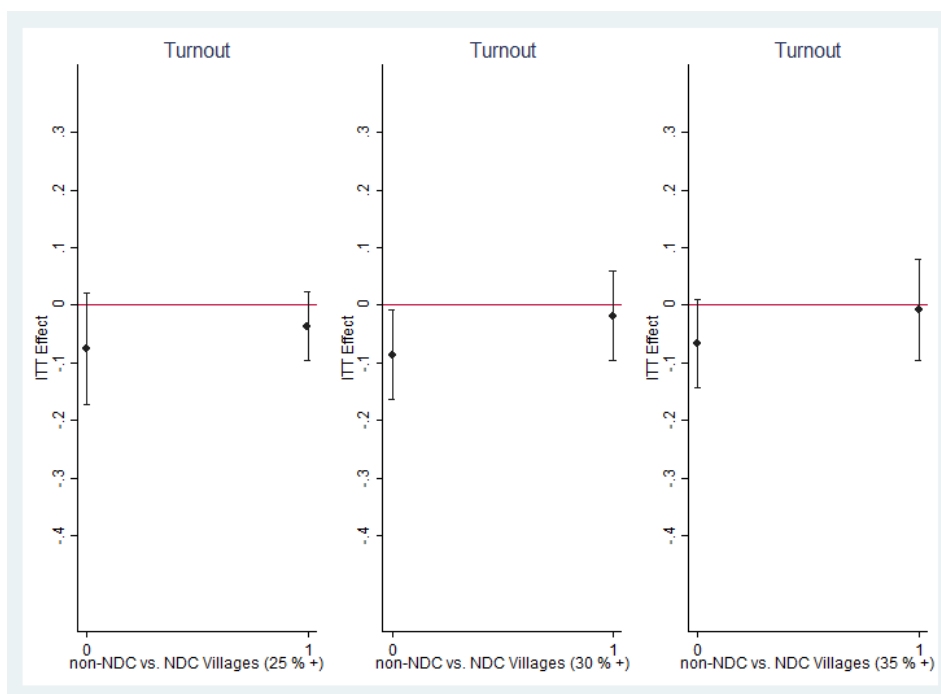


Figure J2. Candidates across non-NDC and NDC Aligned Villages by different cutoffs

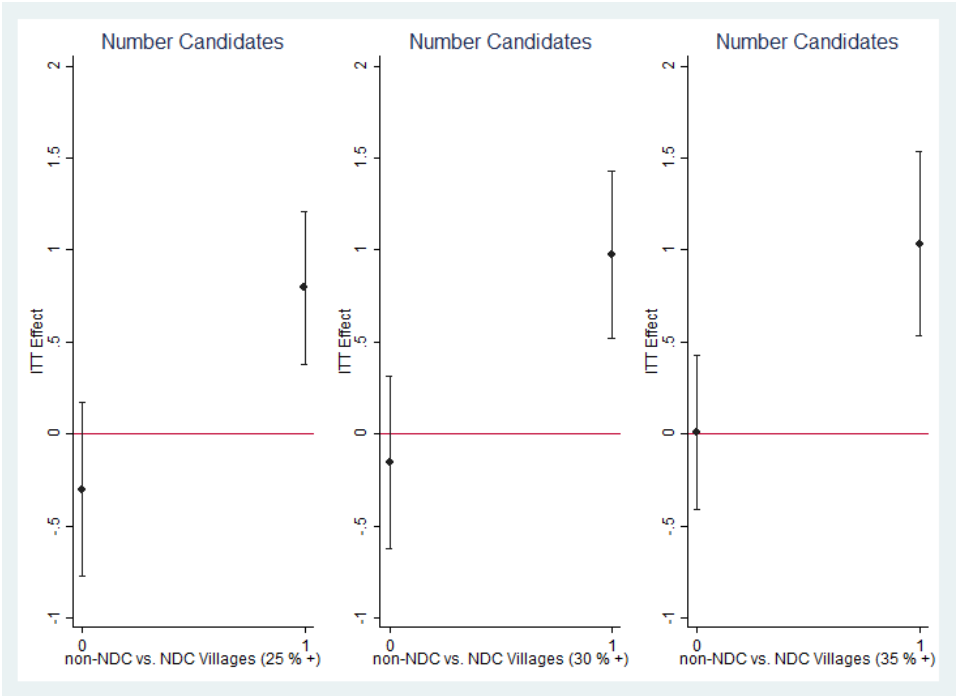


Figure J3. Activity across non-NDC and NDC Aligned Villages by different cutoffs

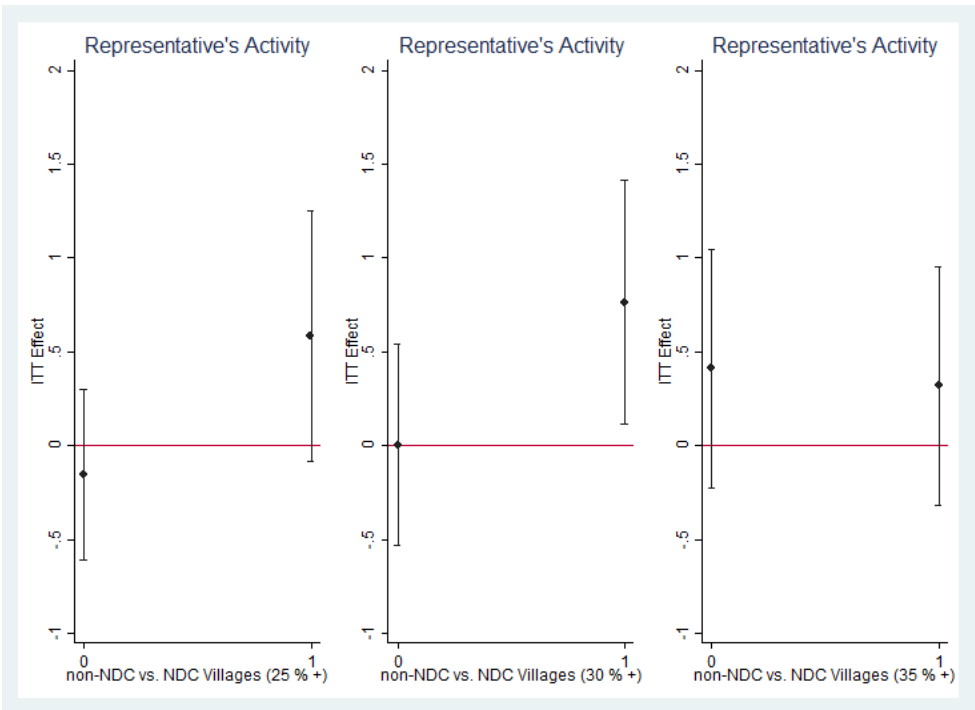


Figure J4. Local Government Projects across non-NDC and NDC Aligned Villages by different cutoffs

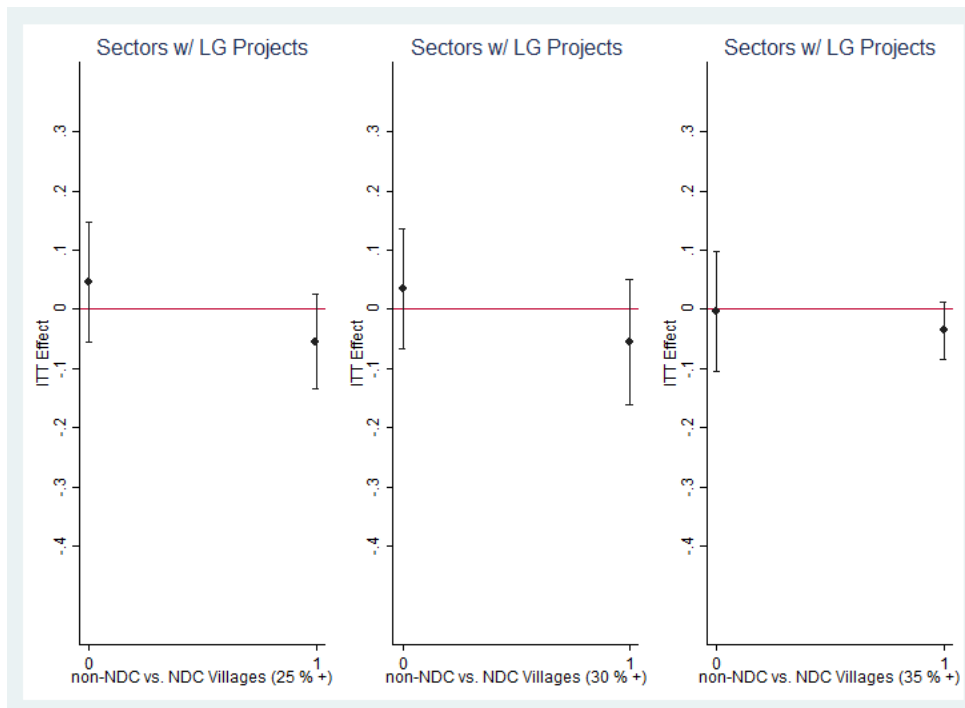


Figure J5. Local Government Projects in THP Sector across non-NDC and NDC Aligned Villages by different cutoffs

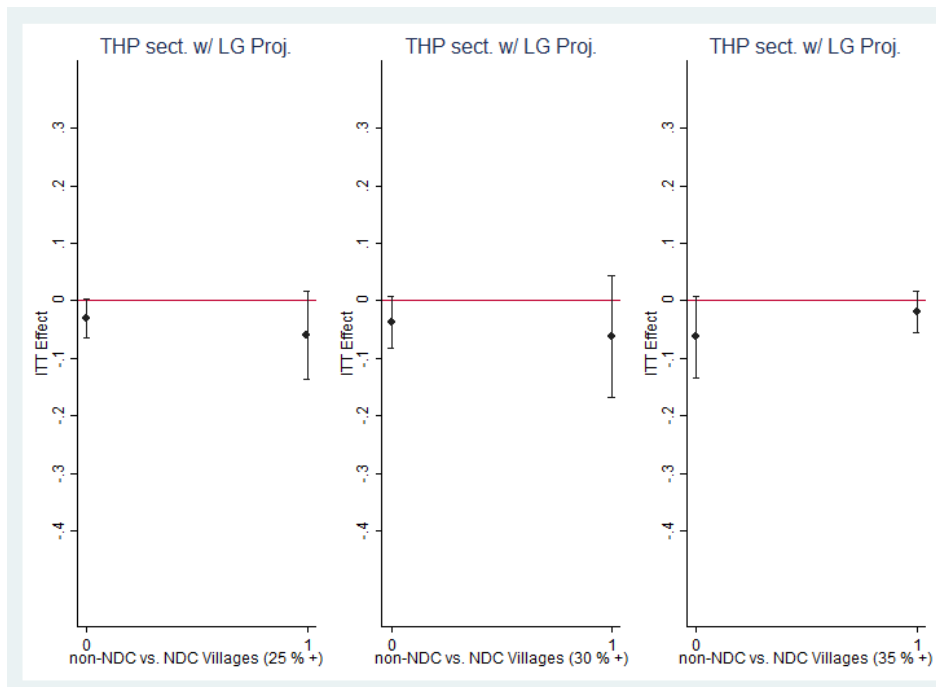
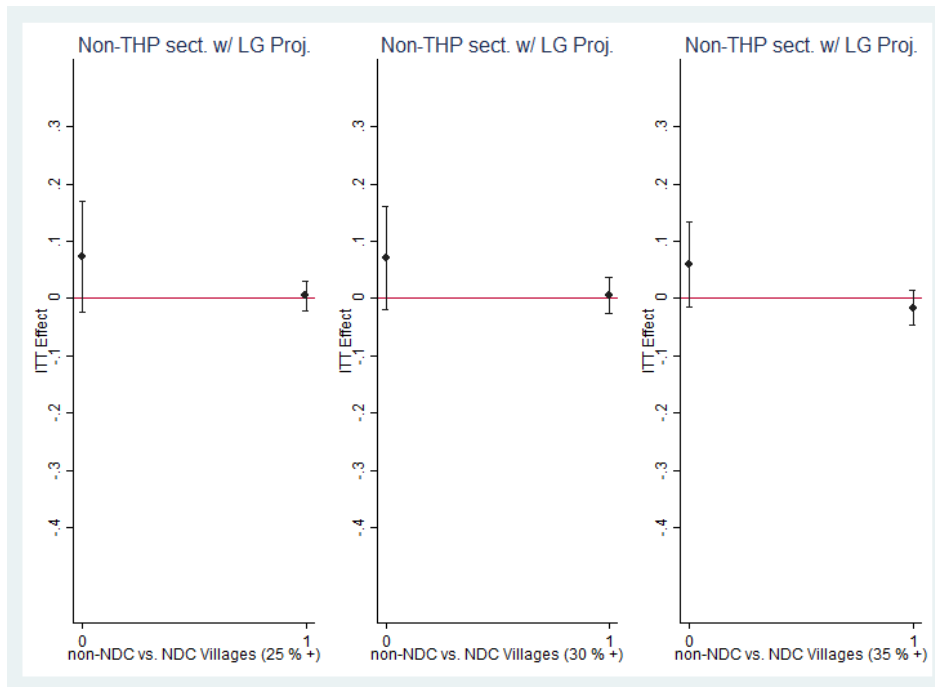


Figure J6. Local Government Projects in non-THP Sector across non-NDC and NDC Aligned Villages by different cutoffs



References

- Humphreys, Macartan, Raúl Sánchez de la Sierra, and Peter van der Windt. 2014. "Social and Economic Impacts of Tuungane: Final Report on the Effects of a Community Driven Reconstruction Program in Eastern Democratic Republic of Congo." In . <https://doi.org/10.23846/ow2232>.
- Mansuri, Ghazala, and Vijayendra Rao. 2013. *Localizing Development: Does Participation Work?* Washington, D.C: World Bank.
- Mosse, David. 2005. *Cultivating Development: An Ethnography of Aid Policy and Practice*. First edition edition. London ; Ann Arbor, MI: Pluto Press.